

# The Effects of Minimum Wage Increases on SNAP Participation: Evidence from a Stacked Difference-in-Differences Design

Haewon Oh\*   Hugo Sant’Anna†   Samyam Shrestha‡   Travis A Smith§

September 15, 2026

## Abstract

Do state minimum-wage increases affect SNAP participation? While earnings gains can lift family income above eligibility thresholds, lost hours and unemployment can push families onto the program. Prior studies on this question disagree on the sign and rely exclusively on state-panel two-way fixed effects (TWFE). Using minimum wage increases of at least \$1 during 2000-2019, we estimate stacked event studies comparing each increase with states whose effective minimum wage remained unchanged during the same period. We show nearly half of the implicit comparisons in the TWFE design are contaminated by already-treated states, and correcting for this reverses their positive finding. Across the full population, we find no meaningful effect on either the extensive margin of participation or the intensive margin of benefit amounts. Restricting to families most likely to be near program eligibility, using a demographic screen built from pre-period data, participation rises modestly but the effect remains statistically insignificant. The response is largest in the first year after an increase, before fading, and pre-trends are flat. Reweighting households by their likelihood of treatment yields a similar pattern.

**JEL Codes:** C23, I38, J38

**Keywords:** Minimum wage, Supplemental Nutrition Assistance Program, food assistance programs

---

\*Auckland University of Technology. Corresponding author.

†University of Alabama at Birmingham.

‡University of Georgia.

§University of Georgia.

# 1 Introduction

The minimum wage and the Supplemental Nutrition Assistance Program (SNAP) are two of the largest policy instruments available to support low-income families in the United States. SNAP grew from 17 million participants and \$15 billion in benefits in fiscal year 2000 to 36 million participants and \$56 billion by 2019, peaking near 48 million in the wake of the Great Recession (U.S. Department of Agriculture, Food and Nutrition Service, 2025). Over the same two decades, most states raised their minimum wages above the federal floor, many repeatedly (U.S. Department of Labor, Wage and Hour Division, 2024). Whether these two policies substitute for each other is a longstanding question in the policy debate. One view holds that a binding wage floor raises earnings past eligibility thresholds and reduces the need for public assistance among low-wage workers (Reich and West, 2015); another holds that job loss and reduced hours at the bottom of the wage distribution can push families onto the program (Clemens and Wither, 2019; Neumark and Shirley, 2022).<sup>1</sup>

This paper estimates the effect of minimum-wage increases on SNAP participation using a stacked event-study design. This approach is well suited to minimum-wage policies because treatment is continuous, states raise their minimum wages repeatedly, and no state remains untreated throughout our sample period. We combine individual data from the Current Population Survey Annual Social and Economic Supplement, including four million individuals across all fifty states and the District of Columbia, from 2000-2019, with the full record of state minimum-wage legislation. Of the 34 within-state increases of at least one dollar over the window, 21 define clean events: increases that can be compared, over a nine-year window, to states whose effective floors stay flat (Figure 1). Stacking these events and saturating with event-by-state and event-by-year fixed effects (Cengiz et al., 2019; Schmidheiny and Siegloch,

---

<sup>1</sup>SNAP’s dual mandate is to increase food purchasing power and to reduce instances of food hardship. Indeed, SNAP is the marginal source of food spending for low-income households (Hastings and Shapiro, 2018), and there is a robust literature showing SNAP reduces food insecurity (for a review, see Smith and Gregory (2023)).

2023) restricts identification to comparisons between newly treated states and genuinely untreated windows. Our analysis focuses on SNAP take-up rather than earnings or poverty, both of which have been studied extensively in the minimum-wage literature.

Population-wide effects on SNAP participation are small and statistically insignificant. The estimated effect on take-up is 0.45 percentage points, with flat pre-trends over four pre-event years, and the 95 percent confidence interval rules out increases larger than 1.1 percentage points per event. A complementary demographic-only screen is estimated in each event’s pre-period and used to select person records likely to live in families with incomes below 130-200% of the Federal Poverty Guidelines (FPG). Because this analysis requires four complete pre-event years, it uses 20 rather than 21 clean events. The pooled OLS estimates are about 0.8 percentage points across the three thresholds and are statistically insignificant. However, the event-study estimates suggest a delayed response, with SNAP participation increasing by roughly 2 percentage points in the first year after a minimum-wage increase. Treatment-propensity-weighted estimates reported in the appendix produce a similar pattern but statistically insignificant pooled effects under a state-cluster bootstrap. A contemporaneous income screen produces larger pooled estimates, but conditions on a post-treatment variable.

These results differ from what the conventional approach finds in the same data. A standard two-way fixed-effects regression of take-up on the minimum-wage level associates each dollar of the wage floor with a statistically significant 0.53-percentage-point rise in SNAP participation, concentrated among less-educated, Black, and Latino families. This difference appears to reflect the limitations of two-way fixed effects in our setting. Because minimum-wage treatment is continuous and recurring, the conventional estimator combines valid comparisons with comparisons that use already-treated states as controls. When treatment effects are dynamic, these comparisons can receive negative weights (Goodman-Bacon, 2021; Chaisemartin and D’Haultfoeuille, 2020; Roth et al., 2023). In our panel, 49.8 percent of the underlying comparison weights are negative. Prior work shows that negative weighting of

this magnitude can substantially distort two-way fixed-effects estimates (Baker, Larcker, and Wang, 2022). The popular cohort-based correctives (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021) assume a binary, absorbing treatment and a reservoir of untreated units, neither of which the minimum wage provides. In this regard, the stacked design matches the structure and nature of the minimum wage increase (Wing, Freedman, and Hollingsworth, 2024).

The take-up estimates above speak only to whether households enroll, not to how much they receive, so we separately examine the intensive margin: whether minimum-wage increases change SNAP benefit amounts even absent a take-up response. Benefit amounts per person and among recipient households alike are statistically unchanged, both in dollars and in logs. This result indicates that minimum-wage increases of the size we study leave SNAP spending unaffected through either the extensive or intensive margin.

This paper makes several contributions to the literature. First, it contributes to the relatively small literature on the minimum wage and public assistance, which has produced mixed findings. Page, Spetz, and Millar (2005) find that higher minimum wages, if anything, *raise* welfare caseloads; Reich and West (2015) estimate that higher minimum wages *reduce* state SNAP caseloads and program expenditures; Snyder, Rinkevich, and Yuan (2019) report *higher* SNAP receipt following increases in high-minimum-wage states; and Sabia and Nguyen (2015) find little net effect on means-tested program participation. These studies exclusively rely on state-panel two-way fixed-effects models. We revisit this question using individual-level microdata and a stacked event-study design that compares treated states with states experiencing no minimum-wage increase over the same event window. In contrast to the positive relationship found in conventional two-way fixed-effects models, we find little evidence of a population-wide effect on SNAP participation. Among individuals predicted to live in lower-income families, the estimates suggest a modest delayed increase in participation, although the pooled effects are statistically insignificant. We also show that nearly half of the comparison weights underlying the conventional two-way fixed-effects estimate are negative.

A much larger literature examines the effects of minimum wages on earnings and family poverty, with mixed findings (Card and Krueger, 1995; Addison and Blackburn, 1999; Dube, 2019; Godoey and Reich, 2021; Neumark and Wascher, 2002; Neumark, Schweitzer, and Wascher, 2005; Burkhauser and Sabia, 2007; Sabia and Burkhauser, 2010; Burkhauser, McNichols, and Sabia, 2025). MaCurdy (2015) shows that minimum-wage earnings gains are spread across the family-income distribution rather than concentrated among poor families. We do not directly examine poverty effects in this paper, but this literature provides an important basis for understanding how minimum-wage changes may affect SNAP participation. Related work examines whether minimum-wage increases reduce employment or hours among low-wage workers (Card and Krueger, 1994; Dube, Lester, and Reich, 2010; Allegretto, Dube, and Reich, 2011; Neumark, Salas, and Wascher, 2014; Meer and West, 2016; Jardim et al., 2022; Manning, 2021). Reductions in employment or hours provide one potential channel through which minimum-wage increases could increase SNAP participation.

Second, this paper contributes to the food-assistance literature. SNAP participation responds to both policy and economic conditions (Ganong and Liebman, 2018; Dickert-Conlin et al., 2021; Jones et al., 2022; Valizadeh, Fischer, and Bryant, 2024), while take-up among eligible households remains incomplete because of information and participation costs (Daponte, Sanders, and Taylor, 1999; Finkelstein and Notowidigdo, 2019). SNAP participation also affects labor supply, food security, and other dimensions of well-being (Hoynes and Schanzenbach, 2012; Gundersen and Ziliak, 2018; East, 2020). Recent studies have examined the relationship between minimum wages and food insecurity, with mixed findings. Minimum-wage increases have been associated with lower severe food insecurity (Nuworsu and Katare, 2025), lower food insecurity in states with earned-income tax credits (Lenhart and Chakraborty, 2025), and *higher* food insecurity in state-panel estimates (Stansel and Wu, 2024). Our analysis contributes to this literature by examining SNAP participation directly using a stacked event-study design that avoids using states with minimum-wage increases within the same event window as controls.

Third, our empirical approach may also be useful in other settings with continuous and recurring policy treatments for which standard staggered-adoption estimators are not well suited (Callaway, Goodman-Bacon, and Sant’Anna, 2024; Chaisemartin and D’Haultfoeulle, 2024). By defining discrete policy events and constructing comparison groups separately for each event, the stacked design accommodates repeated changes in treatment intensity without requiring a permanently untreated group.

The remainder of the paper proceeds as follows. Section 2 sets out the policy background and a conceptual framework for the two opposing channels. Section 3 describes the data and measurement. Section 4 develops the empirical strategy. Section 5 reports the main take-up results, heterogeneity with candidate mechanisms, the reconciliation with conventional estimates, the demographic-screen analysis, and robustness. Section 6 concludes with policy implications.

## **2 Policy Background and Conceptual Model**

### **2.1 SNAP and the Minimum Wage**

SNAP is the largest food assistance program in the United States, and eligibility is largely dependent on family income. Over our sample period, eligibility generally required gross family income below 130 percent of the federal poverty guideline and net income below 100 percent, with benefits declining by roughly thirty cents for each additional dollar of net income. States also varied in their use of broad-based categorical eligibility (e.g., raising the gross income test and asset-test waivers) and reporting/re-certification requirements (e.g., simplified income reporting and differing re-certification periods) (Ganong and Liebman, 2018). Participation tracks both the business cycle and the state policy environment: Dickert-Conlin et al. (2021) attribute roughly half of the 2000-2016 caseload growth to state policy expansions, and program access rules continue to move enrollment at the margin (Jones et al., 2022; Valizadeh, Fischer, and Bryant, 2024). Participation is also incomplete among eligibles:

historically, a third or more of eligible households do not enroll, reflecting information gaps and application frictions (Daponte, Sanders, and Taylor, 1999; Finkelstein and Notowidigdo, 2019).

The minimum-wage environment over the same two decades supplies the policy variation. The federal floor increased from \$5.15 to \$7.25 across 2007-2009 (U.S. Department of Labor, Wage and Hour Division, 2024), and state legislation pushed effective floors above the federal level in most states, often repeatedly and increasingly through indexation. Figure 1 plots every state’s effective minimum wage over 2000-2019 and marks the 21 discrete one-dollar-plus increases our design exploits. Two features of the figure are important for the analysis that follows: the treatment is a continuous dollar amount that moves in steps of heterogeneous size, and no state’s floor stays put for the whole window (i.e., there is no never-treated state).

## 2.2 Conceptual Model

A family’s SNAP status depends on two margins: eligibility, determined by family income relative to the means test, and take-up conditional on eligibility. A minimum-wage increase can affect eligibility in opposite directions. For workers who retain their jobs and hours, a higher minimum wage raises earnings and may move families near the eligibility threshold out of the program. We refer to this as the *exit channel*, which underlies the argument that higher minimum wages may substitute for public assistance (Reich and West, 2015). Alternatively, a minimum-wage increase may reduce employment or hours for some workers (Clemens and Wither, 2019; Jardim et al., 2022; Neumark and Shirley, 2022). The resulting income losses may increase eligibility and SNAP participation, which we refer to as the *entry channel*. The overall effect on SNAP participation therefore depends on the relative importance of these two channels.

Three considerations suggest that the aggregate effect may be small and concentrated among families near the eligibility threshold. Minimum-wage earnings gains tend to be modest and spread across the income distribution, employment effects are generally limited, and

not all eligible families participate in SNAP. We therefore first estimate the population-wide effect on SNAP take-up and then examine demographic subgroups and individuals whose observed characteristics predict that they live in families below the study’s income thresholds. This framework motivates our estimates of population-wide and dynamic effects on SNAP participation, with particular attention to families likely to be near the eligibility threshold.

Consider a continuum of families indexed by  $i$ , each containing a low-wage worker whose hourly wage  $w_i$  is drawn from a distribution  $F(w)$  with support  $[\underline{w}, \bar{w}]$ . Each family’s total income is  $y_i = w_i h_i + \nu_i$ , where  $h_i$  denotes hours worked and  $\nu_i$  denotes non-labor income. The government sets a minimum wage  $\underline{m}$  and administers SNAP with a gross income eligibility threshold  $\bar{y} = \tau \cdot p(n_i)$ , where  $\tau = 1.30$  is the statutory multiplier and  $p(n_i)$  is the U.S. Department of Health and Human Services Federal Poverty Guideline (FPG) for a family of size  $n_i$ . A family is eligible if  $y_i < \bar{y}$  and participates if it is both eligible and clears the take-up hurdle. Following Finkelstein and Notowidigdo (2019), eligible family  $i$  enrolls only if its benefit entitlement  $b_i$  exceeds a privately observed participation cost  $c_i$ , capturing application frictions, stigma, and information gaps. Observed participation is the share of families satisfying both conditions.

These opposing forces can be summarized formally. Letting  $P(\underline{m})$  denote aggregate participation, the total effect is

$$\frac{dP}{d\underline{m}} = - \underbrace{\int_{\{i: y_i \uparrow \bar{y}\}} \mathbb{1}[b_i \geq c_i] dF_i}_{\text{exit channel}} + \underbrace{\int_{\{i: y_i \downarrow \bar{y}\}} \mathbb{1}[b_i \geq c_i] dF_i}_{\text{entry channel}}, \quad (1)$$

where the first integral runs over families whose income crosses  $\bar{y}$  upward and the second over families whose income crosses  $\bar{y}$  downward.

Several considerations suggest that one or both of these effects may be small. First, minimum-wage earnings gains are spread across the family-income distribution rather than concentrated near  $\bar{y}$  (MaCurdy, 2015). Relatively few low-wage workers therefore live in families close enough to the eligibility threshold for an earnings increase to change their

SNAP eligibility. In our data, the median low-wage worker lives in a family with income near three times the poverty line, and only about one in seven has family income below SNAP’s gross-income limit of 130 percent of FPG. For most low-wage workers, an increase in  $w_i$  therefore does not change SNAP eligibility.<sup>2</sup> Second, employment effects of the increases we study are, on the best available evidence, small (Cengiz et al., 2019; Manning, 2021), which limits the families that could enter through lost earnings. Third, for a nontrivial share of eligible families, participation costs exceed expected benefits and the binary indicator  $\mathbb{1}[b_i \geq c_i]$  equals zero (Finkelstein and Notowidigdo, 2019); eligibility changes therefore translate into enrollment changes only imperfectly. As a result, the population-wide effect on SNAP participation may be small even when minimum-wage increases raise wages for affected workers. This does not rule out larger responses among families near the eligibility threshold. In particular, income losses associated with reductions in employment or hours may increase SNAP participation for some families, potentially with a delay. Our empirical analysis therefore examines both the population-wide response and the response among individuals likely to live in lower-income families.

## 3 Data

### 3.1 Sample and Outcome

Our analysis combines individual-level data from the Current Population Survey Annual Social and Economic Supplement (CPS-ASEC) (Flood et al., 2024) with state minimum wage series (Vaghul and Zipperer, 2016) for years 2000 through 2019.

Minimum-wage policy directly affects workers, whereas SNAP receipt is measured at

---

<sup>2</sup>Computed on the sample of low-wage workers (earnings below two-thirds of the national median hourly wage, positive reported wages; 835,472 person-years, weighted by the ASEC sampling weight): family income-to-poverty has a median of 3.1 and an interquartile range of 1.9 to 6.0, and 13.8 percent of these workers live in families below 130 percent of poverty against 27.4 percent below 200 percent. The comparable share below 130 percent is 18.9 percent in the full CPS-ASEC population and 16.3 percent among low-wage householders alone, so the statistic does not turn on counting teenagers and secondary earners in higher-income families.

the household level. The CPS records whether anyone in the household received SNAP during the prior twelve months, and we assign this household-level indicator to every sampled person. Our unrestricted main sample therefore retains children, adults age 65 and older, and people with disabilities. It contains 4,012,269 person records across the 50 states and the District of Columbia over income years 2000-2019. The population-weighted coefficient therefore measures the change in the share of individuals living in a household that reports SNAP receipt. We do not restrict the main sample based on income or wages because these variables may themselves respond to minimum-wage changes. Instead, we examine demographic subgroups with different degrees of labor-market exposure in Section 5.2. As a complementary analysis, Section 5.5 uses a demographic-only model, estimated in each event’s pre-period, to select individual records likely to live in families below specified income thresholds. The model uses each row’s observed demographics, which can change over time, but does not use income, wages, hours, or employment as predictors.

Our primary outcome is therefore a person-level indicator of whether the individual’s household reports receiving SNAP. This measure allows us to use the individual CPS-ASEC records directly, but SNAP receipt is subject to substantial reporting error in household surveys. Comparisons of survey responses with administrative records show substantial underreporting of SNAP receipt in the CPS (Bollinger and David, 1997; Meyer and Mittag, 2019; Meyer, Mittag, and Goerge, 2022). Using administrative records from 19 states matched to the 2017-2019 CPS Food Security Supplement, Gregory and O’Connor (2026) estimate that 47.3 percent of administrative SNAP participants do not report receipt. Although their analysis uses a different CPS supplement and period, it provides additional evidence that SNAP underreporting in the CPS is substantial. Underreporting does not necessarily attenuate the estimates toward zero. If reporting behavior varies with income, employment, program contact, or minimum-wage exposure, the direction of the resulting bias is ambiguous. We therefore interpret our estimates as effects on reported SNAP receipt in the CPS-ASEC. Future work using administrative participation data could assess whether similar patterns

appear in program records.

## 3.2 Summary Statistics

Table 1 summarizes the full estimating sample of 4,012,269 individuals in the CPS-ASEC over income years 2000-2019, covering all fifty states and the District of Columbia. The average individual is 37.2 years old, 51.0 percent are women, and 34.3 percent have less than a high school education. Reported SNAP receipt averages 10.3 percent, and the poverty rate is 13.9 percent. In terms of family income relative to the federal poverty guideline, 6.2 percent of the sample lives below 50 percent of the guideline, 12.7 percent between 50 and 130 percent, 12.7 percent between 130 and 200 percent, and 68.5 percent at or above twice the guideline. The FPG ratio is defined as family income divided by the HHS Federal Poverty Guideline for the family’s size. We use this ratio to describe the sample’s income distribution and to define the income benchmarks used in Section 5.5. The poverty rate is instead measured against the Census official poverty threshold, which adjusts for family size and number of children. This last fact matters for what follows. A population-wide design asks a minimum-wage increase to move an outcome whose base rate is one in ten, among a sample more than two-thirds of which sits above twice the poverty guideline, far from the eligibility threshold, which is why we also examine demographic subgroups (Section 5.2) and the demographic-screen sample (Section 5.5).

Figure 2 plots annual reported SNAP receipt for the full population and the demographic subgroups used below. Each series is the ASEC-weighted share of individuals in that group whose household reports receipt, computed from the unstacked sample. The full-population share rises from 6.3 percent in 2000 to a peak of 13.9 percent in 2013, then falls to 9.8 percent in 2019.

Table 1 does not report a treated-versus-control comparison, and deliberately so. Every state’s effective minimum wage rose at some point during 2000-2019, if only through the 2007-2009 federal steps (U.S. Department of Labor, Wage and Hour Division, 2024), so there

is no state that is untreated in levels. More importantly, the stacked design does not partition states into a treated group and a control group once and for all. It assigns treatment *within each event window*: the state raising its floor by at least one dollar is treated for that window, and every state recording no such increase inside the same window is a control there. A state treated in one window therefore serves as a control in others, and the composition of both groups changes from event to event (Section 4.1). A fixed two-column split would describe a comparison the estimates never make.

## 4 Empirical Strategy

Our empirical strategy is motivated by three features of the minimum wage as a treatment. First, treatment is *continuous* because the minimum wage varies in dollar amounts rather than switching from untreated to treated. Second, treatment is *recurring* because states may raise their minimum wages multiple times during the sample period. Third, there is *no never-treated state*, since every state’s effective minimum wage increases at some point during our sample period, including through changes in the federal minimum wage (see Figure 1). These features complicate the use of conventional two-way fixed-effects models and cohort-based staggered-adoption estimators (Roth et al., 2023; Chaisemartin and D’Haultfœuille, 2023). We therefore define discrete minimum-wage events and compare states experiencing a qualifying increase with states that experience no qualifying increase over the same event window. We first describe this stacked event-study design and then present the conventional two-way fixed-effects specifications used for comparison with the prior literature.

### 4.1 Stacked Event Studies

Our preferred specification uses a stacked event-study design, following Cengiz et al. (2019), Schmidheiny and Siegloch (2023), and Wing, Freedman, and Hollingsworth (2024). We define an event as a within-state minimum-wage increase of at least \$1 and use a nine-year window

spanning four years before and four years after the increase. The \$1 threshold excludes many smaller increases associated with inflation indexation and federal pass-through, while the four-year pre-period allows us to examine pre-treatment trends. For each event, the state experiencing the qualifying minimum-wage increase is the treated state, and the comparison group consists of states with no increase of at least \$1 during the same event window. We drop an event if the treated state experiences another qualifying increase within its own window. We then stack the event-specific samples and include event-by-state and event-by-year fixed effects. This structure restricts comparisons to treated and comparison states within the same event window and avoids using states with qualifying increases during that window as controls. We estimate

$$Y_{iste} = \sum_{k \neq -1} \theta_k \mathbf{1}[K_{ste} = k] \cdot \text{Treated}_{se} + X'_{iste} \gamma + \phi_{se} + \psi_{te} + \varepsilon_{iste}, \quad (2)$$

where  $e$  indexes events,  $K_{ste}$  denotes event time, and the year immediately before the increase ( $k = -1$ ) is the omitted reference category. The coefficients  $\theta_k$  trace the dynamic response of SNAP take-up before and after the increase, and their pre-period values ( $k = -4, -3, -2$ ) serve as a test of parallel trends. A static version, which replaces the event-time indicators with a single post-increase indicator, summarizes the results as one average effect. Twenty-one clean events satisfy this definition over 2000-2019. Table A1 lists their states, event years, observed windows, and minimum-wage changes.<sup>3</sup>

The population-wide participation specification includes the following controls in  $X$ . At the individual level we control for age, sex, marital status, household size, an indicator for a child under age six in the household, and race and ethnicity. At the state level we add two time-varying controls: the Federal Housing Finance Agency house-price index and a Bartik-style labor-demand shock, built as a shift-share of national industry employment

---

<sup>3</sup>One of the twenty-one, Alaska's 2003 increase, is the earliest in the sample, so its pre-period runs three years rather than four: the  $k = -4$  lead is estimated on the other twenty events, while  $k = -3$  and  $k = -2$  use all twenty-one. Because the specification is saturated with event-by-state and event-by-year fixed effects, the unbalanced lead does not affect the static post-versus-pre contrast that we report as the headline estimate.

growth applied to each state’s base-period industry mix (Bartik, 1991; Goldsmith-Pinkham, Sorkin, and Swift, 2020; Burkhauser, McNichols, and Sabia, 2025).<sup>4</sup> We do not include the state SNAP-policy index used to capture variation in program access rules (Ganong and Liebman, 2018) in the baseline specification. Changes in state SNAP policies may occur alongside minimum-wage changes, and conditioning on contemporaneous policy changes could absorb part of the policy environment associated with a minimum-wage increase. We therefore assess sensitivity to the inclusion of the SNAP-policy index separately. Adding the index changes the estimated SNAP effect by only 0.0003, well within its standard error (Table A2).

Because the \$1 cutoff is a modeling choice, we assess sensitivity by redefining qualifying events using thresholds of \$0.50 and \$0.70 in addition to the baseline \$1.00 threshold. Lower thresholds admit smaller minimum-wage increases and may also allow comparison states to experience increases just below the relevant cutoff within the event window. We therefore report both the estimated effects and pre-treatment trend tests for each threshold in Section 5.4. Standard errors are clustered by state throughout.

We do not condition the whole-population analysis on contemporaneous family income. Section 5.5 instead uses a demographic prediction model estimated from pre-event income-target indicators to identify individuals likely to live in lower-income families. Income and labor-market variables are excluded from the prediction model. Because the screen is applied using demographics observed in each year, some of which may change after treatment, we refer to it as a demographic screen rather than a predetermined sample.

---

<sup>4</sup>We use annual-average private-sector employment from the BLS Quarterly Census of Employment and Wages, aggregated to two-digit NAICS sectors. The control is  $B_{st} = \sum_j s_{sj,2000}(E_{jt}^{US}/E_{j,2000}^{US} - 1)$ , where  $s_{sj,2000}$  is sector  $j$ ’s share of state  $s$ ’s observed private employment across these sectors in 2000 and  $E_{jt}^{US}$  sums that sector’s employment across the 50 states and the District of Columbia, including state  $s$ . Shares remain fixed at 2000 values, so the index measures predicted cumulative proportional employment growth relative to 2000. We merge it by state and income year, one year before the ASEC survey year.

## 4.2 Conventional Specifications for Comparison

The specification that recurs throughout the applied minimum-wage literature is a two-way fixed-effects regression of an individual outcome on the minimum wage,

$$Y_{ist} = \beta D_{st} + X'_{ist}\gamma + W'_{st}\delta + \phi_s + \psi_t + \varepsilon_{ist}, \quad (3)$$

where  $Y_{ist}$  is SNAP take-up for individual  $i$  in state  $s$  and year  $t$ ,  $D_{st}$  is the minimum-wage treatment,  $\phi_s$  and  $\psi_t$  are state and year fixed effects, and  $X$  and  $W$  collect the same individual and state controls as Equation (2). We estimate Equation (3) in two forms, familiar from the prior literature: one with  $D_{st}$  equal to the level of the effective floor in dollars, which yields a per-dollar association, and one with  $D_{st} = \Delta \ln MW_{st}$ , the log-change treatment of Burkhauser, McNichols, and Sabia (2025). Standard errors are clustered by state.

We report these estimates for comparison with earlier studies that use conventional two-way fixed-effects specifications. When treatment timing varies across states and treatment effects differ across states or over time, the two-way fixed-effects coefficient can combine comparisons with different implicit weights, including comparisons in which previously treated states serve as controls for later-treated states (Goodman-Bacon, 2021; Chaisemartin and D'Haultfœuille, 2020). Some of these comparisons may receive negative weights, which can complicate interpretation of the resulting coefficient (Baker, Larcker, and Wang, 2022). Applying the Chaisemartin and D'Haultfœuille (2020) decomposition to our panel, we find that 508 of the 1,020 underlying state-by-year comparisons, or 49.8 percent, receive negative weight. The negative weights sum to  $-5.26$ , while the positive weights sum to  $6.26$ . Because no state remains untreated throughout the full sample period, the conventional specification relies on differences in the timing and magnitude of minimum-wage changes across states. Section 5.3 compares estimates from this specification with those from the stacked event-study design.

We also do not apply standard cohort-based staggered-adoption estimators (Sun and

Abraham, 2021; Callaway and Sant’Anna, 2021; Borusyak, Jaravel, and Spiess, 2024; Gardner et al., 2025). These methods are generally formulated around a discrete treatment-adoption event, often with treatment remaining in place after adoption. Minimum-wage policy differs because treatment intensity is continuous and states may experience multiple increases during the sample period. Our stacked design instead defines each qualifying minimum-wage increase as a separate event and constructs the comparison group separately within each event window.

## 5 Results

### 5.1 Population-wide Average Effects on SNAP Participation

Table 2, column (1), reports the population-wide estimate. The effect of a one-dollar-plus minimum-wage increase on the share of people living in a household that reports SNAP receipt is 0.0045 (standard error 0.0031,  $p = 0.15$ ), or approximately 0.45 percentage points. The estimate is statistically insignificant at conventional levels. The joint test of the pre-treatment coefficients does not reject the null of no differential pre-trends ( $p = 0.40$ ).

The 95 percent confidence interval ranges from approximately  $-0.2$  to  $+1.1$  percentage points per event. The upper bound is about one-tenth of the sample SNAP participation rate of 10.3 percent. The point estimate is also smaller than the approximately 0.7 percentage-point response implied by the conventional per-dollar estimate in Section 5.3 for the average qualifying increase of \$1.29. Although the confidence interval includes effects of this magnitude, it rules out increases larger than approximately 1.1 percentage points and reductions larger than approximately 0.2 percentage points. The estimate therefore provides little evidence of a large population-wide change in SNAP participation.

Figure 3 plots the dynamic coefficients. The pre-increase coefficients ( $k = -4$  through  $-2$ ) are small and statistically indistinguishable from zero, providing no evidence of differential pre-trends. The post-treatment estimates remain statistically insignificant in most event years, although the coefficient at event year  $+3$  is positive and statistically significant (0.0099,

standard error 0.0045,  $p = 0.03$ ).

The positive estimate at event year +3 is consistent with the delayed pattern observed in the demographic-screen analysis in Section 5.5. Previous studies also document gradual adjustments in SNAP participation following changes in economic conditions and program policies. Ganong and Liebman (2018) find that local unemployment rates and SNAP policy changes affect participation with lags of up to three years, while Melvin and Smith (2022) document lagged responses following the rollout of Electronic Benefit Transfer (EBT).

## 5.2 Heterogeneity and candidate mechanisms

Table 3 reports estimates for demographic subgroups commonly examined in the minimum-wage literature, and Figure 4 presents the corresponding event-study estimates. Most subgroup estimates are statistically insignificant. SNAP participation increases significantly among younger less-educated individuals, with no evidence of differential pre-trends, while the estimate for the broader working-age group is positive and marginally significant. The Latino estimate is also statistically significant, but its event study rejects the null of no differential pre-trends. Estimates for less-educated, Black, and immigrant (foreign-born) individuals are statistically insignificant, with no evidence of differential pre-trends.

SNAP take-up among *younger less-educated* family members rises by 2.3 percentage points (0.0227,  $p = 0.024$ ), with no evidence of differential pre-trends ( $p = 0.69$ ). This pattern is consistent with the entry channel discussed in Section 2.2. Younger and less-educated workers may be more exposed to reductions in employment or hours following minimum-wage increases (Clemens and Wither, 2019; Neumark and Shirley, 2022), which could increase SNAP eligibility for some families.

The estimate for Latino individuals is also statistically significant (0.0140,  $p = 0.025$ ), but the corresponding event study rejects the null of no differential pre-trends ( $p = 0.02$ ). SNAP participation among Latino individuals in treated and comparison states was already evolving differently before the qualifying minimum-wage increases. We therefore do not interpret the

post-treatment estimate as evidence of a treatment effect.

Under the conventional level specification, the estimated association is statistically significant at the one-percent level for the full sample and for the working-age, less-educated, and Black subgroups, and at the five-percent level for Latino individuals (column 3 of Table 3). In the stacked design, the younger less-educated subgroup has the largest estimate among the groups with no evidence of differential pre-trends. The estimate for the broader working-age group is smaller and marginally significant (0.0052,  $p < 0.10$ ), while the estimate for the less-educated subgroup is statistically insignificant. The Latino estimate remains similar in magnitude (0.0140) but is accompanied by evidence of differential pre-trends and therefore is not interpreted as a treatment effect. Section 5.3 compares these patterns with those obtained from the conventional specifications.

### 5.3 Reconciling with Conventional Estimates

Columns (2) and (3) of Table 2 report estimates from the conventional specifications. In the level linear probability model, each additional dollar of the effective minimum wage is associated with a 0.0053 increase in SNAP take-up (standard error 0.0016,  $p = 0.002$ ). The Burkhauser, McNichols, and Sabia (2025) log-change specification also produces a positive estimate (0.0312,  $p = 0.19$ ), although it is statistically insignificant at conventional levels. Thus, the conventional level specification suggests a positive association between minimum wages and SNAP participation, in contrast to the statistically insignificant population-wide estimate from the stacked design.

The difference between the conventional and stacked estimates is consistent with the weighting issues discussed in Section 4.2. In the conventional two-way fixed-effects specification, 49.8 percent of the underlying comparison weights are negative, and previously treated states may contribute to comparisons with states experiencing later minimum-wage increases. The subgroup results also differ across specifications. The positive associations for Black and less-educated individuals in the conventional specification are statistically insignificant in the

stacked design, while the Latino estimate remains similar in magnitude but is accompanied by evidence of differential pre-trends. Section 5.4 further examines whether the stacked estimates are sensitive to the threshold used to define qualifying minimum-wage events.

## 5.4 Robustness Checks

Because the \$1 event threshold is a modeling choice, Table 4 examines sensitivity to alternative thresholds. At the baseline \$1.00 threshold, the estimated effect is 0.0045 ( $p = 0.15$ , 21 events, pre-trend  $p = 0.40$ ). At the \$0.70 threshold, the estimate is  $-0.0018$  ( $p = 0.53$ , 43 events), while at the \$0.50 threshold it is 0.0081 ( $p = 0.02$ , 19 events). The estimates at the \$1.00 and \$0.70 thresholds are small and statistically insignificant.

The \$0.50 specification also rejects the null of no differential pre-trends ( $p = 0.02$ ), which limits a causal interpretation of the positive post-treatment estimate. This lower threshold includes smaller minimum-wage changes, including some associated with inflation indexation, and allows comparison states to experience increases of just under \$0.50 within the event window. In contrast, the \$0.70 and \$1.00 specifications produce statistically insignificant estimates and do not reject the pre-trend tests. We therefore place greater weight on the results from these higher thresholds.

Two additional analyses assess the robustness of the population-wide estimate. First, collapsing the data to state-by-year cells and using the whole-population SNAP participation share as the outcome produces a similarly small and statistically insignificant estimate ( $+0.0030$ ,  $p = 0.37$ ; Table A2, Panel A). Second, adding the state SNAP-policy index of Ganong and Liebman (2018) changes the estimated take-up effect by 0.0003, which is small relative to its standard error (Table A2, Panel B).

## 5.5 Demographic Screens and SNAP Take-Up

The whole-population design may mask responses among people likely to have low family income. For each event, we therefore fit a person-weighted logit in its four pre-treatment

years for whether a sampled person lives in a family with an FPG ratio below 1.30, 1.85, or 2.00. The 1.30 cutoff is the statutory gross-income limit under regular SNAP rules. The 1.85 cutoff reflects the BBCE-adjusted limit, as most states participating in BBCE raised their gross-income threshold to 185 percent of the FPG. The 2.00 benchmark is the broad upper cutoff retained for continuity. Predictors are a quadratic in age; sex; family size; the number, ages, and presence of children; marital and single-mother status; race and ethnicity; nativity and citizenship; and education. We apply the fitted model to the demographics observed in each row throughout the event window and choose an event-specific cutoff that makes the ASEC-weighted selected share equal the observed pre-period share below the target. Earnings, hours, employment, and income do not enter. Because row demographics can change over time, this is a demographic screen rather than a predetermined-sample design. Requiring four complete pre-event years excludes Alaska’s 2003 event, leaving 20 clean events. Unlike the fixed one-condition splits in Section 5.2, each screen combines several characteristics, is re-estimated by event, and is calibrated to the event’s pre-period income-target prevalence. The closest subgroup is less-educated person records, but its weighted Jaccard overlap with the screens is only 0.36-0.44. The screens select person records; they neither identify SNAP assistance units nor determine statutory eligibility.

Figure A1 compares annual reported SNAP receipt in the full event-stack population and the three demographic-screen samples. Both panels use the same event windows and ASEC weights. These descriptive shares summarize the samples used below; their year-to-year movements also reflect changes in the events contributing observations.

Table 5 reports the main ASEC-weighted OLS estimates. Outcome regressions include house-price and Bartik labor-demand controls, event-by-state fixed effects, and event-by-year fixed effects, but no additional individual demographic controls. The pooled estimates are +0.0083, +0.0079, and +0.0079 for the FPG 1.30, 1.85, and 2.00 screens, respectively, and none are statistically significant. The corresponding event studies in Figure 5 show no evidence of differential pre-trends (joint  $p = 0.90, 0.96, \text{ and } 0.94$ ) and suggest a similar

delayed pattern across the three screens. At event year +1, the estimates are +2.3, +1.9, and +1.8 percentage points ( $p = 0.030, 0.039$ , and  $0.058$ ), while the other post-treatment coefficients are statistically insignificant at the five-percent level.

Table A3 and figure A2 report treatment-propensity weighting as an appendix robustness check. After screening, a separate additive event-specific logit of treated-state status uses age, sex, marital status, household size, presence of a child under six, race and ethnicity, and nativity observed across the full event window to reweight control records toward treated records. The pooled IPW estimates are about 1.0 percentage point at all three benchmarks and remain statistically insignificant under a 300-replication state-cluster bootstrap ( $p = 0.19, 0.15$ , and  $0.14$ ); the dynamic pattern is similar to OLS. The bootstrap re-estimates treatment weights while holding the demographic screen fixed. We also retain the contemporaneous-income comparison in Table A4 and figure A3 of the appendix. That screen selects person records using family income that the minimum wage can change; its larger pooled estimates remain subject to post-treatment selection. A simulated statutory-eligibility design would additionally require state-year program rules and measures of SNAP assistance units, countable monthly income, deductions, and assets that the present ASEC construction does not fully recover; we therefore leave that exercise for future work.

## 5.6 The Intensive Margin: SNAP Benefit Amounts

The participation indicator counts households, not dollars. The fiscal question behind the employer-subsidy debate is about dollars: if higher wage floors raise SNAP spending, that can happen through new enrollments, through larger benefits for continuing recipients, or both, and a null on take-up leaves the second channel open. The benefit formula also gives the intensive margin a prediction of its own. SNAP benefits fall by about thirty cents for each additional dollar of net income, so a household that keeps its job and hours, stays enrolled, and receives a raise should see its benefit shrink even though its participation status does not change. A pure earnings gain among recipients would therefore lower benefit amounts, while

a pure entry response would raise them.

We estimate the stacked design of Equation (2) with the household SNAP benefit amount for the prior twelve months, deflated to 2019 dollars, as the outcome. As with the receipt indicator, the household amount is copied to each person record. We report two versions. The unconditional version uses all person records and includes zeros for non-recipient households, so it combines the extensive and intensive margins and measures the change in SNAP dollars per person; its sample mean is \$372 per person-year. The conditional version retains person records in recipient households (mean benefit \$3,622 per year) and isolates benefit amounts among participants. Because receipt can itself respond to treatment, the conditional sample is selected on a post-treatment outcome, and we read those estimates as descriptive of the recipient population rather than as causal. Each version is estimated in dollars and in logs.

Table 6 reports the static estimates. None is statistically distinguishable from zero. The unconditional dollar effect is +\$9.9 per person-year (standard error \$19.0,  $p = 0.60$ ). Its 95 percent confidence interval runs from about  $-\$27$  to  $+\$47$ , or from  $-7$  to  $+13$  percent of the sample mean, so the design rules out changes in SNAP dollars per person larger than about an eighth of the average. The unconditional log estimate is  $+0.029$  ( $p = 0.27$ ). Conditional on receipt, the dollar estimate is  $-\$45.5$  (standard error \$80.1,  $p = 0.57$ ), about 1.3 percent of the recipient mean, and the log estimate is  $+0.003$  ( $p = 0.91$ ). The benefit-taper prediction of a decline among continuing recipients is not rejected, but the interval is too wide for the point estimate to carry that reading.

Figure 6 plots the corresponding event studies. All four panels have flat pre-trends: the joint tests on the leads return  $p$ -values of 0.92, 0.54, 0.50, and 0.25. The post-event coefficients are small and imprecise, and none is significant at five percent. The unconditional dollar coefficients alternate in sign around zero from  $k = 0$  to  $+4$ . The unconditional log series is the only one with a hint of the delayed pattern seen in take-up: its largest coefficient is at event year  $+3$  ( $+0.074$ ,  $p = 0.08$ ), the same year in which take-up peaks in Figure 3. The conditional dollar series is negative from event year  $+1$  onward, but the three leads are negative and of

similar size ( $-\$140$  to  $-\$215$ ), so the post-event coefficients mirror the pre-period rather than mark a decline: the reference year  $k = -1$  is a local high. No post-event coefficient is significant at five percent, and the largest, at  $+3$  ( $-\$257$ ,  $p = 0.09$ ), is about seven percent of the recipient mean. Read together with Section 5.1, the intensive margin moves no more than the extensive one: minimum-wage increases of the size we study leave SNAP dollars per person, and benefit amounts among recipients, statistically unchanged.

## 6 Discussion and Conclusions

This study examines whether state minimum-wage increases affect SNAP participation. Using CPS-ASEC microdata from 2000 to 2019 and a stacked event-study design that avoids comparisons between newly treated and already-treated states, we find little evidence that minimum-wage increases alter SNAP participation at the population level. Benefit amounts do not move either: SNAP dollars per person and benefit levels among recipient households are statistically unchanged after an increase, with flat pre-trends in every specification. The SNAP findings are consistent across demographic subgroups, alternative treatment definitions, and multiple robustness checks. In contrast, conventional two-way fixed-effects estimates suggest a positive relationship between minimum wages and SNAP participation. This difference is consistent with the weighting issues that arise in conventional two-way fixed-effects models when treatment timing and effects vary across states.

The findings have implications for ongoing debates regarding the interaction between labor market policies and the social safety net. Some argue that higher minimum wages may increase reliance on SNAP by reducing employment opportunities, while others contend that higher wages should reduce program participation by raising household income. Our results provide little evidence that either mechanism produces large population-wide changes in SNAP participation. At the same time, the population average masks some heterogeneity. One explanation is that earnings increases are modest relative to household income and are

not concentrated among households near SNAP eligibility thresholds. In addition, incomplete program take-up among eligible households may further weaken the connection between changes in eligibility and observed enrollment. Selecting person records whose observed demographics predict residence in a family below three FPG benchmarks produces a delayed and modest take-up pattern. The pooled OLS estimates remain statistically insignificant and the treatment-propensity-weighted appendix estimates produce the same conclusion under the state-cluster bootstrap. We therefore interpret the dynamic pattern as a suggestive refinement of the population-wide null rather than a reversal of it.

These results suggest that minimum-wage policy should not be expected to generate large changes in SNAP participation at the population level. SNAP participation may instead respond more directly to program-specific policies, including eligibility rules, benefit levels, and administrative access. More broadly, although labor-market policies and safety-net programs jointly shape household economic well-being, our findings indicate that the population-wide relationship between minimum-wage increases and reported SNAP participation is limited.

Several limitations should be noted. First, SNAP receipt is substantially underreported in federal surveys, and the direction of bias need not be simple attenuation if reporting behavior responds to income, employment, or program contact. Administrative state or county participation outcomes would therefore provide an important validation of the CPS-based null. Second, the CPS provides state identifiers but does not allow us to examine local variation in minimum-wage policies or program administration. Third, the demographic screens are income-benchmark proxies rather than simulated statutory eligibility. Reconstructing the latter would require state-year program rules and measures of SNAP assistance units, countable monthly income, deductions, and assets that the current ASEC construction does not fully observe. Future research using administrative data and a dedicated rule simulation could provide more precise estimates of participation among families near eligibility thresholds. Within these limitations, the evidence indicates that recent state minimum-wage increases had little effect on population-wide reported SNAP participation, while the estimates for

individuals likely to live in lower-income families suggest a modest and potentially delayed response.

## References

- Addison, J. T. and M. L. Blackburn (1999). “Minimum Wages and Poverty”. In: *Industrial and Labor Relations Review* 52.3, pp. 393–409. DOI: 10.1177/001979399905200302.
- Allegretto, S. A., A. Dube, and M. Reich (2011). “Do Minimum Wages Really Reduce Teen Employment? Accounting for Heterogeneity and Selectivity in State Panel Data”. In: *Industrial Relations: A Journal of Economy and Society* 50.2, pp. 205–240. DOI: 10.1111/j.1468-232X.2011.00634.x.
- Baker, A. C., D. F. Larcker, and C. C. Y. Wang (2022). “How Much Should We Trust Staggered Difference-in-Differences Estimates?” In: *Journal of Financial Economics* 144.2, pp. 370–395. DOI: 10.1016/j.jfineco.2022.01.004.
- Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* Kalamazoo, MI: W.E. Upjohn Institute for Employment Research. DOI: 10.17848/9780585223940.
- Bollinger, C. R. and M. H. David (1997). “Modeling Discrete Choice with Response Error: Food Stamp Participation”. In: *Journal of the American Statistical Association* 92.439, pp. 827–835. DOI: 10.1080/01621459.1997.10474038.
- Borusyak, K., X. Jaravel, and J. Spiess (2024). “Revisiting Event-Study Designs: Robust and Efficient Estimation”. In: *Review of Economic Studies* 91.6, pp. 3253–3285. DOI: 10.1093/restud/rdae007.
- Burkhauser, R. V., D. McNichols, and J. J. Sabia (2025). “Minimum Wages and Poverty: New Evidence from Dynamic Difference-in-Differences Estimates”. In: *Review of Economics and Statistics*. Forthcoming. DOI: 10.1162/rest\_a\_01590.
- Burkhauser, R. V. and J. J. Sabia (2007). “The Effectiveness of Minimum-Wage Increases in Reducing Poverty: Past, Present, and Future”. In: *Contemporary Economic Policy* 25.2, pp. 262–281. DOI: 10.1111/j.1465-7287.2006.00045.x.

- Callaway, B., A. Goodman-Bacon, and P. H. C. Sant’Anna (2024). *Difference-in-Differences with a Continuous Treatment*. NBER Working Paper 32117. Conditionally accepted, American Economic Review. National Bureau of Economic Research. DOI: 10.3386/w32117.
- Callaway, B. and P. H. C. Sant’Anna (2021). “Difference-in-Differences with Multiple Time Periods”. In: *Journal of Econometrics* 225.2, pp. 200–230. DOI: 10.1016/j.jeconom.2020.12.001.
- Card, D. and A. B. Krueger (1994). “Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania”. In: *American Economic Review* 84.4, pp. 772–793.
- Card, D. and A. B. Krueger (1995). *Myth and Measurement: The New Economics of the Minimum Wage*. Princeton, NJ: Princeton University Press.
- Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). “The Effect of Minimum Wages on Low-Wage Jobs”. In: *Quarterly Journal of Economics* 134.3, pp. 1405–1454. DOI: 10.1093/qje/qjz014.
- Chaisemartin, C. de and X. D’Haultfœuille (2020). “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects”. In: *American Economic Review* 110.9, pp. 2964–2996. DOI: 10.1257/aer.20181169.
- Chaisemartin, C. de and X. D’Haultfœuille (2023). “Two-Way Fixed Effects and Differences-in-Differences with Heterogeneous Treatment Effects: A Survey”. In: *Econometrics Journal* 26.3, pp. C1–C30. DOI: 10.1093/ectj/utac017.
- Chaisemartin, C. de and X. D’Haultfœuille (2024). “Difference-in-Differences Estimators of Intertemporal Treatment Effects”. In: *Review of Economics and Statistics*. Advance access; volume and issue pending, pp. 1–18. DOI: 10.1162/rest\_a\_01414.
- Clemens, J. and M. Wither (2019). “The Minimum Wage and the Great Recession: Evidence of Effects on the Employment and Income Trajectories of Low-Skilled Workers”. In: *Journal of Public Economics* 170, pp. 53–67. DOI: 10.1016/j.jpubeco.2019.01.004.

- Daponte, B. O., S. Sanders, and L. Taylor (1999). “Why Do Low-Income Households Not Use Food Stamps? Evidence from an Experiment”. In: *Journal of Human Resources* 34.3, pp. 612–628. DOI: 10.2307/146382.
- Dickert-Conlin, S., K. Fitzpatrick, B. Stacy, and L. Tiehen (2021). “The Downs and Ups of the SNAP Caseload: What Matters?” In: *Applied Economic Perspectives and Policy* 43.3, pp. 1026–1050. DOI: 10.1002/aepp.13076.
- Dube, A. (2019). “Minimum Wages and the Distribution of Family Incomes”. In: *American Economic Journal: Applied Economics* 11.4, pp. 268–304. DOI: 10.1257/app.20170085.
- Dube, A., T. W. Lester, and M. Reich (2010). “Minimum Wage Effects Across State Borders: Estimates Using Contiguous Counties”. In: *Review of Economics and Statistics* 92.4, pp. 945–964. DOI: 10.1162/REST\_a\_00039.
- East, C. N. (2020). “The Effect of Food Stamps on Children’s Health: Evidence from Immigrants’ Changing Eligibility”. In: *Journal of Human Resources* 55.2, pp. 387–427. DOI: 10.3368/jhr.55.3.0916–8197R2.
- Finkelstein, A. and M. J. Notowidigdo (2019). “Take-Up and Targeting: Experimental Evidence from SNAP”. In: *The Quarterly Journal of Economics* 134.3, pp. 1505–1556. DOI: 10.1093/qje/qjz013.
- Flood, S., M. King, R. Rodgers, S. Ruggles, J. R. Warren, D. Backman, A. Chen, G. Cooper, S. Richards, M. Schouweiler, and M. Westberry (2024). *IPUMS CPS: Version 12.0 [dataset]*. Minneapolis, MN. DOI: 10.18128/D030.V12.0.
- Ganong, P. and J. B. Liebman (2018). “The Decline, Rebound, and Further Rise in SNAP Enrollment: Disentangling Business Cycle Fluctuations and Policy Changes”. In: *American Economic Journal: Economic Policy* 10.4, pp. 153–176. DOI: 10.1257/pol.20140016.
- Gardner, J., N. Thakral, L. T. Tô, and L. Yap (2025). *Two-Stage Differences in Differences*. Working paper, arXiv:2207.05943. DOI: 10.48550/arXiv.2207.05943.

- Godoe, A. and M. Reich (2021). “Are Minimum Wage Effects Greater in Low-Wage Areas?” In: *Industrial Relations: A Journal of Economy and Society* 60.1, pp. 36–83. DOI: 10.1111/irel.12267.
- Goldsmith-Pinkham, P., I. Sorkin, and H. Swift (2020). “Bartik Instruments: What, When, Why, and How”. In: *American Economic Review* 110.8, pp. 2586–2624. DOI: 10.1257/aer.20181047.
- Goodman-Bacon, A. (2021). “Difference-in-Differences with Variation in Treatment Timing”. In: *Journal of Econometrics* 225.2, pp. 254–277. DOI: 10.1016/j.jeconom.2021.03.014.
- Gregory, C. A. and K. O’Connor (2026). *The Supplemental Nutrition Assistance Program and Food Security: Insights From Administrative Records*. Economic Research Report ERR-355. U.S. Department of Agriculture, Economic Research Service. URL: <https://www.ers.usda.gov/publications/115069>.
- Gundersen, C. and J. P. Ziliak (2018). “Food Insecurity Research in the United States: Where We Have Been and Where We Need to Go”. In: *Applied Economic Perspectives and Policy* 40.1, pp. 119–135. DOI: 10.1093/aepp/ppx058.
- Hastings, J. and J. M. Shapiro (2018). “How Are SNAP Benefits Spent? Evidence from a Retail Panel”. In: *American Economic Review* 108.12, pp. 3493–3540. DOI: 10.1257/aer.20170866.
- Hoynes, H. W. and D. W. Schanzenbach (2012). “Work Incentives and the Food Stamp Program”. In: *Journal of Public Economics* 96.1-2, pp. 151–162. DOI: 10.1016/j.jpubeco.2011.08.006.
- Jardim, E., M. C. Long, R. Plotnick, E. van Inwegen, J. Vigdor, and H. Wething (2022). “Minimum-Wage Increases and Low-Wage Employment: Evidence from Seattle”. In: *American Economic Journal: Economic Policy* 14.2, pp. 263–314. DOI: 10.1257/pol.20180578.
- Jones, J. W., C. Courtemanche, A. Denteh, J. Marton, and R. Tchernis (2022). “Do State Supplemental Nutrition Assistance Program Policies Influence Program Participation

- among Seniors?” In: *Applied Economic Perspectives and Policy* 44.2, pp. 591–608. DOI: 10.1002/aepp.13231.
- Lenhart, O. and K. Chakraborty (2025). “Hungry no More? The Joint Impact of Minimum Wages and the Earned Income Tax Credit on Food Insecurity”. In: *Health Economics* 34.9, pp. 1509–1517. DOI: 10.1002/hec.70001.
- MaCurdy, T. (2015). “How Effective Is the Minimum Wage at Supporting the Poor?” In: *Journal of Political Economy* 123.2, pp. 497–545. DOI: 10.1086/679626.
- Manning, A. (2021). “The Elusive Employment Effect of the Minimum Wage”. In: *Journal of Economic Perspectives* 35.1, pp. 3–26. DOI: 10.1257/jep.35.1.3.
- Meer, J. and J. West (2016). “Effects of the Minimum Wage on Employment Dynamics”. In: *Journal of Human Resources* 51.2, pp. 500–522. DOI: 10.3368/jhr.51.2.0414-6298R1.
- Melvin, S. and T. A. Smith (2022). “From paper to plastic: How the transition to EBT affected SNAP enrollment”. In: *Economics Letters* 220, p. 110852.
- Meyer, B. D. and N. Mittag (2019). “Using Linked Survey and Administrative Data to Better Measure Income: Implications for Poverty, Program Effectiveness, and Holes in the Safety Net”. In: *American Economic Journal: Applied Economics* 11.2, pp. 176–204. DOI: 10.1257/app.20170478.
- Meyer, B. D., N. Mittag, and R. M. Goerge (2022). “Errors in Survey Reporting and Imputation and Their Effects on Estimates of Food Stamp Program Participation”. In: *Journal of Human Resources* 57.5, pp. 1605–1644. DOI: 10.3368/jhr.58.1.0818-9704R2.
- Neumark, D., J. M. I. Salas, and W. Wascher (2014). “Revisiting the Minimum Wage–Employment Debate: Throwing Out the Baby with the Bathwater?” In: *ILR Review* 67.Supplement, pp. 608–648. DOI: 10.1177/00197939140670S307.
- Neumark, D., M. Schweitzer, and W. Wascher (2005). “The Effects of Minimum Wages on the Distribution of Family Incomes: A Nonparametric Analysis”. In: *Journal of Human Resources* 40.4, pp. 867–894. DOI: 10.3368/jhr.XL.4.867.

- Neumark, D. and P. Shirley (2022). “Myth or Measurement: What Does the New Minimum Wage Research Say about Minimum Wages and Job Loss in the United States?” In: *Industrial Relations: A Journal of Economy and Society* 61.4, pp. 384–417. DOI: 10.1111/irel.12306.
- Neumark, D. and W. Wascher (2002). “Do Minimum Wages Fight Poverty?” In: *Economic Inquiry* 40.3, pp. 315–333. DOI: 10.1093/ei/40.3.315.
- Nuworsu, E. K. and B. Katare (2025). “How Does Increasing the Minimum Wage Affect Household Food Insecurity?” In: *Journal of the Agricultural and Applied Economics Association* 4.3, pp. 391–404. DOI: 10.1002/jaa2.70019.
- Page, M. E., J. Spetz, and J. Millar (2005). “Does the Minimum Wage Affect Welfare Caseloads?” In: *Journal of Policy Analysis and Management* 24.2, pp. 273–295. DOI: 10.1002/pam.20090.
- Reich, M. and R. West (2015). “The Effects of Minimum Wages on Food Stamp Enrollment and Expenditures”. In: *Industrial Relations: A Journal of Economy and Society* 54.4, pp. 668–694. DOI: 10.1111/irel.12110.
- Roth, J., P. H. C. Sant’Anna, A. Bilinski, and J. Poe (2023). “What’s Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature”. In: *Journal of Econometrics* 235.2, pp. 2218–2244. DOI: 10.1016/j.jeconom.2023.03.008.
- Sabia, J. J. and R. V. Burkhauser (2010). “Minimum Wages and Poverty: Will a \$9.50 Federal Minimum Wage Really Help the Working Poor?” In: *Southern Economic Journal* 76.3, pp. 592–623. DOI: 10.4284/sej.2010.76.3.592.
- Sabia, J. J. and T. T. Nguyen (2015). *The Effects of Minimum Wage Increases on Means-Tested Government Assistance*. Tech. rep. Washington, DC: Employment Policies Institute. URL: <https://epionline.org/studies/the-effects-of-minimum-wage-increases-on-means-tested-government-assistance/>.

- Schmidheiny, K. and S. Sieglöcher (2023). “On Event Studies and Distributed-Lag Models in Two-Way Fixed Effects Designs: Identification, Equivalence, and Generalization”. In: *Journal of Applied Econometrics* 38.5, pp. 695–713.
- Smith, T. A. and C. A. Gregory (2023). “Food insecurity in the United States: measurement, economic modeling, and food assistance effectiveness”. In: *Annual Review of Resource Economics* 15.1, pp. 279–303.
- Snyder, T., S. Rinkevich, and W. Yuan (2019). “Do Minimum Wage Increases Affect SNAP Benefits?” In: *The B.E. Journal of Economic Analysis & Policy* 19.2, p. 20180045. DOI: 10.1515/bejeap-2018-0045.
- Stansel, D. and F. Wu (2024). “Economic Freedom, the Minimum Wage, and Food Insecurity”. In: *Applied Economic Perspectives and Policy* 46.3, pp. 1127–1150. DOI: 10.1002/aepp.13438.
- Sun, L. and S. Abraham (2021). “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects”. In: *Journal of Econometrics* 225.2, pp. 175–199. DOI: 10.1016/j.jeconom.2020.09.006.
- U.S. Department of Agriculture, Food and Nutrition Service (2025). *SNAP Data Tables: National Level Annual Summary, Participation and Costs, 1969–2024*. <https://www.fns.usda.gov/pd/supplemental-nutrition-assistance-program-snap>.
- U.S. Department of Labor, Wage and Hour Division (2024). *History of Changes to the Minimum Wage Law*. <https://www.dol.gov/agencies/whd/minimum-wage/history>.
- Vaghul, K. and B. Zipperer (2016). *Historical State and Sub-State Minimum Wage Data*. Working Paper. Washington, DC: Washington Center for Equitable Growth. URL: <https://equitablegrowth.org/working-papers/historical-state-and-sub-state-minimum-wage-data/>.
- Valizadeh, P., B. L. Fischer, and H. L. Bryant (2024). “SNAP Enrollment Cycles: New Insights from Heterogeneous Panel Models with Cross-Sectional Dependence”. In: *American Journal of Agricultural Economics* 106.1, pp. 354–381. DOI: 10.1111/ajae.12390.

Wing, C., S. M. Freedman, and A. Hollingsworth (2024). *Stacked Difference-in-Differences*.  
NBER Working Paper 32054. National Bureau of Economic Research. DOI: 10.3386/  
w32054.

**Table 1:** Summary Statistics, Full Estimating Sample

|                                                 | Mean      |
|-------------------------------------------------|-----------|
| <i>Demographics</i>                             |           |
| Age                                             | 37.2      |
| Female (%)                                      | 51.0      |
| Married (%)                                     | 41.2      |
| Less than high school (%)                       | 34.3      |
| Black (%)                                       | 12.2      |
| Latino (%)                                      | 16.3      |
| Child under 16 (%)                              | 21.3      |
| <i>Labor &amp; income</i>                       |           |
| Mean FPG ratio                                  | 4.25      |
| Poverty rate (%)                                | 13.9      |
| SNAP receipt (%)                                | 10.3      |
| <i>Income relative to poverty guideline (%)</i> |           |
| FPG < 0.50                                      | 6.2       |
| FPG 0.50-1.29                                   | 12.7      |
| FPG 1.30-1.99                                   | 12.7      |
| FPG $\geq$ 2.00                                 | 68.5      |
| Individuals                                     | 4,012,269 |
| States                                          | 51        |
| Years                                           | 20        |

*Notes:* The sample is all individuals in the CPS-ASEC over income years 2000-2019, weighted by ASEC sampling weights (ASECWT): 4,012,269 individuals across all fifty states and the District of Columbia. Cells are weighted sample means; indicator variables are reported as percentages. The table does not report separate treated and control groups because treatment status is defined within each event window. A state is treated in a window when it experiences a qualifying minimum-wage increase and may serve as a control in another window if it experiences no qualifying increase during that window. Thus, the stacked design does not produce a fixed treated-control partition for the full sample. The FPG ratio is family income divided by the HHS Federal Poverty Guideline for the family's size; the same ratio defines the income benchmarks used in Section 5.5. The bracket cut points correspond to the 1.30 gross-income benchmark and the 2.00 broad-based categorical eligibility benchmark. The reported poverty rate is based separately on the Census official poverty threshold, which varies by family size and number of children. The income panel partitions the sample into four mutually exclusive FPG brackets; the FPG ratio is defined for 99.97% of weighted records.

**Table 2:** Minimum Wage and SNAP Take-Up: Main Estimates

|                      | (1)                                       | (2)                   | (3)                |
|----------------------|-------------------------------------------|-----------------------|--------------------|
| SNAP take-up         | 0.0045<br>(0.0031)                        | 0.0053***<br>(0.0016) | 0.0312<br>(0.0235) |
| Estimator            | Stacked ES                                | LPM                   | LPM                |
| Treatment            | Clean \$1 events                          | MW level (\$)         | $\Delta \ln$ MW    |
| Estimand             | Post-increase ATT                         | Per \$1               | Per log point      |
| Demographic controls | Yes                                       | Yes                   | Yes                |
| FE                   | Event $\times$ state, event $\times$ year | State, year           | State, year        |
| Pre-trend $p$        | 0.40                                      | -                     | -                  |
| Observations         | 27,367,646                                | 4,012,269             | 3,794,000          |
| Clean events         | 21                                        | -                     | -                  |

*Notes:* The dependent variable is the household SNAP-receipt indicator, referring to the prior twelve months, copied to each person record. The sample is all individuals in the CPS-ASEC, income years 2000-2019. Column (1) is the paper’s preferred design: the static clean stacked event study, estimated on the individual-level micro data, with treatment defined as a within-state effective-minimum-wage increase of at least \$1.00 (21 clean events), event-by-state and event-by-year fixed effects, population-weighted; its observation count is the stacked person-year sample, in which an individual record appears once for each clean event window it serves in as treated or control, and the pre-trend row reports the joint  $p$ -value on the event-time leads ( $k = -2, -3, -4$ ). No collapsing to state-year cells is involved in any column. Columns (2) and (3) are the conventional two-way fixed-effects specifications, shown for comparison with the prior literature: column (2) is a level linear probability model (LPM) with the effective minimum wage in dollars, and column (3) is an LPM with the Burkhauser, McNichols, and Sabia (2025) log-change treatment,  $\Delta \ln$  MW; both are estimated on the individual micro data with state and year fixed effects. All columns include individual demographic controls (age, sex, race, marital status, household size, and a young-children indicator) and house-price and Bartik labor-demand controls. Standard errors (in parentheses) are clustered by state (51 states). \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Table 3:** Heterogeneity in the SNAP Take-Up Effect, by Subgroup

|                          | (1)<br>Clean stacked | (2)<br>Pre-trend $p$ | (3)<br>Naive MW level (\$) |
|--------------------------|----------------------|----------------------|----------------------------|
| All individuals          | 0.0045<br>(0.0031)   | 0.40                 | 0.0053***<br>(0.0016)      |
| Working age              | 0.0052*<br>(0.0030)  | 0.58                 | 0.0052***<br>(0.0016)      |
| Less educated            | 0.0050<br>(0.0049)   | 0.69                 | 0.0072***<br>(0.0023)      |
| Younger, less educated   | 0.0227**<br>(0.0098) | 0.69                 | 0.0090*<br>(0.0045)        |
| Black                    | 0.0068<br>(0.0072)   | 0.20                 | 0.0100***<br>(0.0036)      |
| Latino/Hispanic          | 0.0140**<br>(0.0061) | 0.02                 | 0.0105**<br>(0.0046)       |
| Immigrant (foreign-born) | 0.0042<br>(0.0044)   | 0.62                 | 0.0035<br>(0.0033)         |

*Notes:* The dependent variable is the household SNAP-receipt indicator copied to each person record. Each row re-estimates the take-up effect on the indicated person-record subgroup. The sample is all individuals in the CPS-ASEC, income years 2000-2019. Column (1) is the clean stacked event study (\$1.00 events, 21 clean events); column (2) reports the joint pre-trend  $p$ -value (event time  $k = -2, -3, -4$ ) for the clean estimate; column (3) is the conventional level minimum-wage (\$) linear probability model with state and year fixed effects, shown for comparison. All specifications include individual demographic controls and house-price and Bartik labor-demand controls. Standard errors (in parentheses) are clustered by state (51 states). \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Table 4:** Robustness: The Event-Threshold Ladder

| Event threshold | Estimate | Std. err. | $p$ -value | Pre-trend $p$ | Clean events |
|-----------------|----------|-----------|------------|---------------|--------------|
| \$0.50          | 0.0081** | (0.0033)  | 0.018      | 0.024         | 19           |
| \$0.70          | -0.0018  | (0.0029)  | 0.535      | 0.327         | 43           |
| \$1.00          | 0.0045   | (0.0031)  | 0.152      | 0.396         | 21           |

*Notes:* The dependent variable is the household SNAP-receipt indicator copied to each person record. The sample is all individuals in the CPS-ASEC, income years 2000-2019. Each row re-estimates the static clean stacked event study of SNAP take-up with the event threshold, the minimum within-state effective-minimum-wage increase that qualifies as a treatment event, re-defined at \$0.50, \$0.70, and \$1.00; the pre-trend column reports the joint  $p$ -value on the event-time leads ( $k = -2, -3, -4$ ), and the final column the number of clean events at each threshold. All specifications are estimated on the individual micro data with individual demographic controls, house-price and Bartik labor-demand controls, and event-by-state and event-by-year fixed effects. Standard errors (in parentheses) are clustered by state (51 states). \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Table 5:** Minimum Wage and SNAP Take-Up: Demographic-Screen Estimates

| FPG benchmark | Stacked ES<br>(1)  |
|---------------|--------------------|
| FPG < 1.30    | 0.0083<br>(0.0071) |
| FPG < 1.85    | 0.0079<br>(0.0060) |
| FPG < 2.00    | 0.0079<br>(0.0058) |

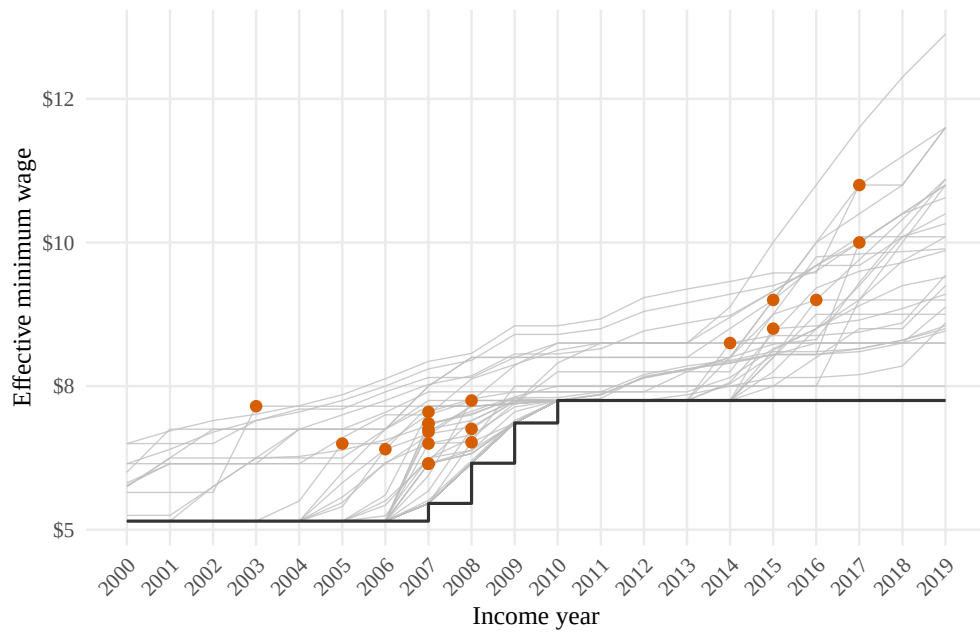
*Notes:* Static stacked event-study estimates of the pooled post-versus-pre effect of a  $\geq$  \$1 minimum-wage increase on the household SNAP-receipt indicator copied to each selected person record. For each event, the demographic screen is estimated in four pre-treatment years and applied to observed row demographics throughout the event window; a prevalence-matched cutoff selects the same ASEC-weighted share as the pre-period income target. FPG denotes family income relative to the HHS Federal Poverty Guidelines. The 1.30 cutoff is the regular-rule gross-income benchmark, 1.85 is a BBCE-adjusted benchmark, and 2.00 is the broad upper benchmark retained for continuity. OLS uses ASEC weights. Outcome regressions include house-price and Bartik controls and event-by-state and event-by-year fixed effects, but no additional individual demographic controls. Standard errors are clustered by state. Requiring four complete pre-event years excludes Alaska's 2003 event, so all estimates use 20 events. The selected person records form demographic proxies for low-income status, not determinations of family-level statutory SNAP eligibility.

**Table 6:** Effect on SNAP Benefit Amounts, Unconditional and Conditional on Receipt

|                                    | Benefit (2019\$)<br>(1) | Log benefit<br>(2) |
|------------------------------------|-------------------------|--------------------|
| Unconditional (all person records) | 9.9<br>(19.0)           | 0.0291<br>(0.0259) |
| Conditional on receipt             | -45.5<br>(80.1)         | 0.0035<br>(0.0294) |

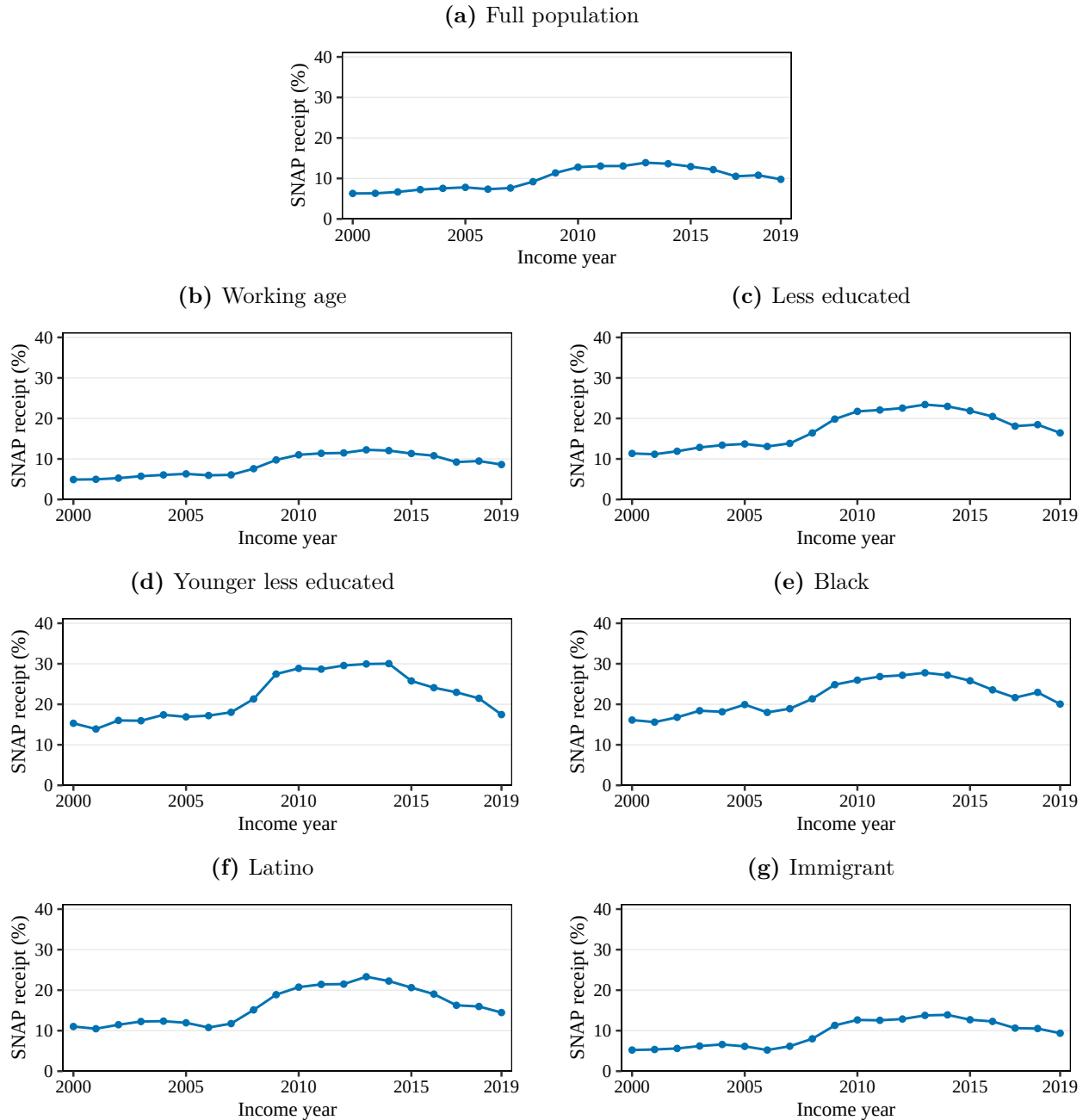
*Notes:* Static clean stacked event-study estimates (\$1.00 events, 21 clean events) of the effect of a minimum-wage increase on the household SNAP benefit amount, deflated to 2019 dollars by the CPI-U. Column (1) measures the benefit in dollars per year; column (2) measures the log benefit. As with the receipt indicator, the household benefit amount is copied to each person record. The *unconditional* rows include all person records, zeros included, and combine the extensive and intensive margins; the estimating sample contains 27,367,646 stacked person-years. The *conditional* rows retain person records in households with positive SNAP receipt (mean household benefit \$3,622 per year; 2,897,185 stacked person-years). Because SNAP receipt can respond to treatment, the conditional estimates are composition-aware rather than causal. All specifications use ASEC weights, house-price and Bartik labor-demand controls, and event-by-state and event-by-year fixed effects, with no individual demographic controls; joint pre-trend  $p$ -values exceed 0.24 in every cell, and Figure 6 plots the dynamic coefficients. Standard errors (in parentheses) are clustered by state (51 states).  $*p < 0.10$ ,  $**p < 0.05$ ,  $***p < 0.01$ .

**Figure 1:** State Minimum-Wage Variation and the Clean Events, 2000-2019



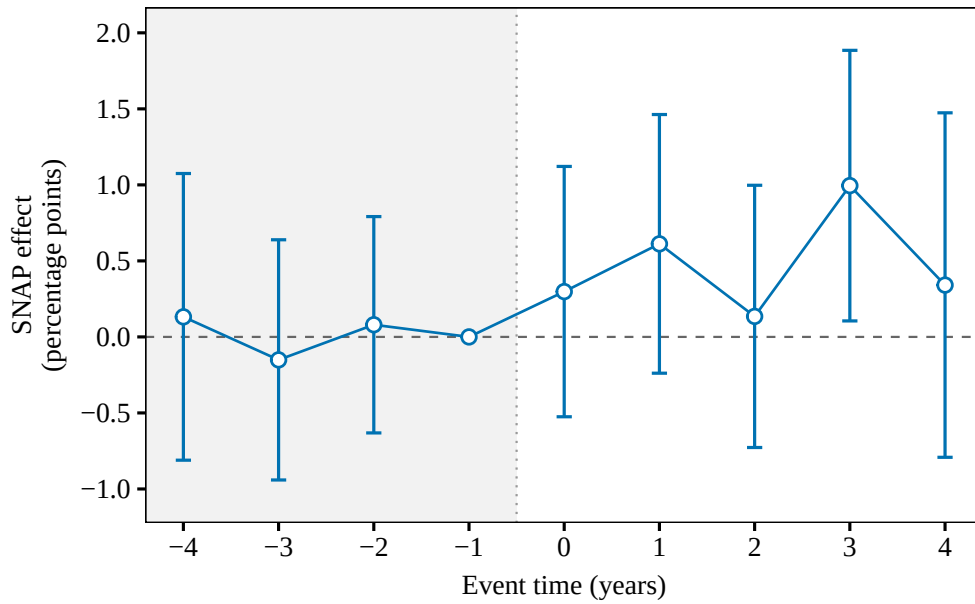
*Notes:* Each grey line is one state's effective minimum wage (the higher of the state and federal statutory rates) by income year, 2000-2019 (Vaghul and Zipperer, 2016). The dark step line is the federal floor, which rose from \$5.15 to \$7.25 across 2007-2009. Orange points mark the 21 clean events used in the stacked design: within-state year-over-year increases of at least \$1.00 with no other qualifying increase by the same state within four years on either side.

**Figure 2:** Reported SNAP Receipt over the Sample Period, by Demographic Group



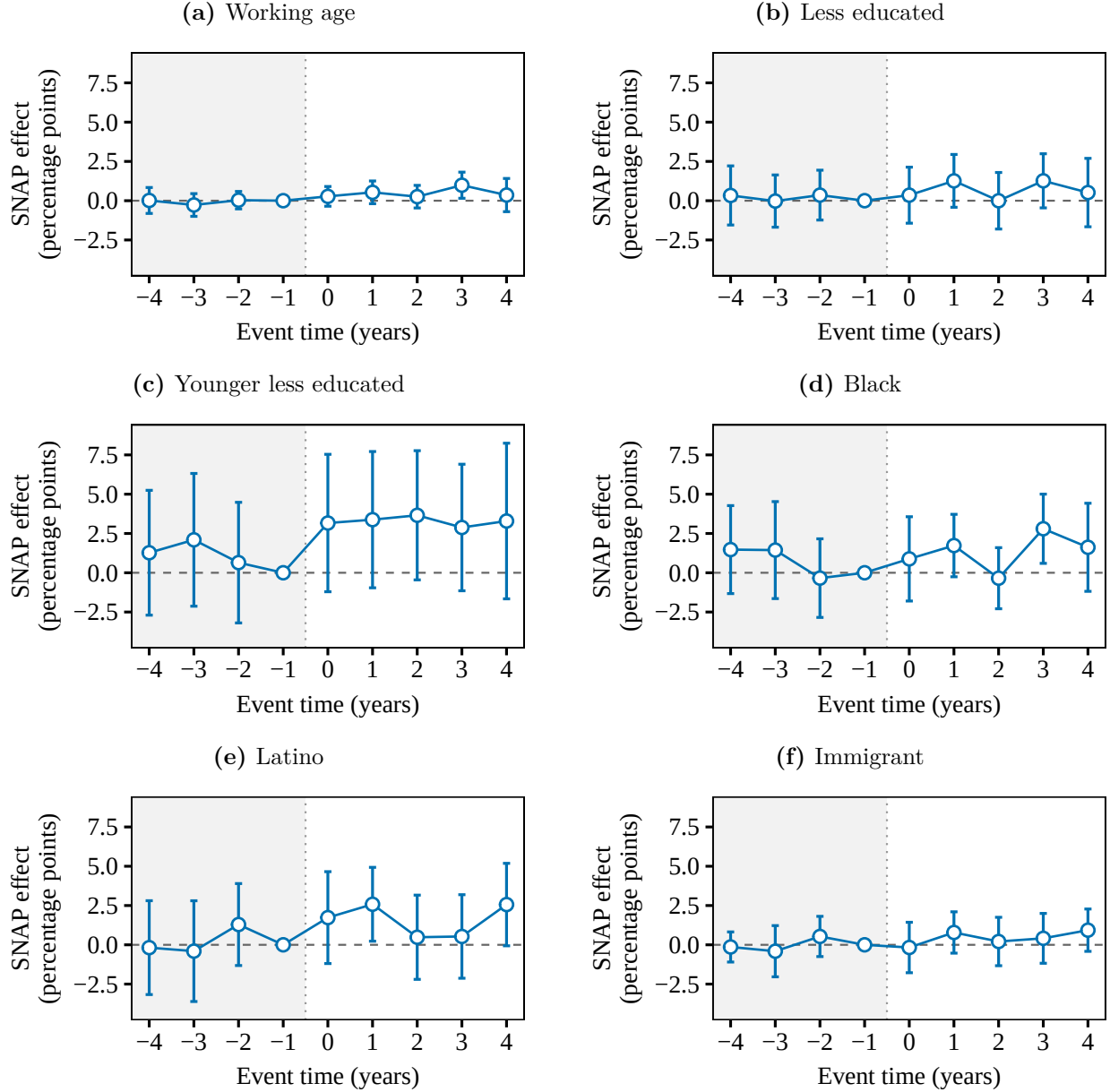
*Notes:* Annual ASEC-weighted shares of individuals whose household reports SNAP receipt, using the unstacked CPS-ASEC sample over income years 2000–2019. Each series uses the indicated group’s own population as its denominator; the vertical scale is common across panels. Working age denotes ages 18–64; less educated denotes less than a high school education, including children; younger less educated restricts that group to ages 18–30. Black denotes non-Hispanic Black individuals; Latino denotes Hispanic individuals; immigrants are foreign-born. Groups overlap. These are descriptive survey-reported receipt rates, not administrative participant counts or treatment-effect estimates.

**Figure 3:** Clean Stacked Event Study of SNAP Take-Up



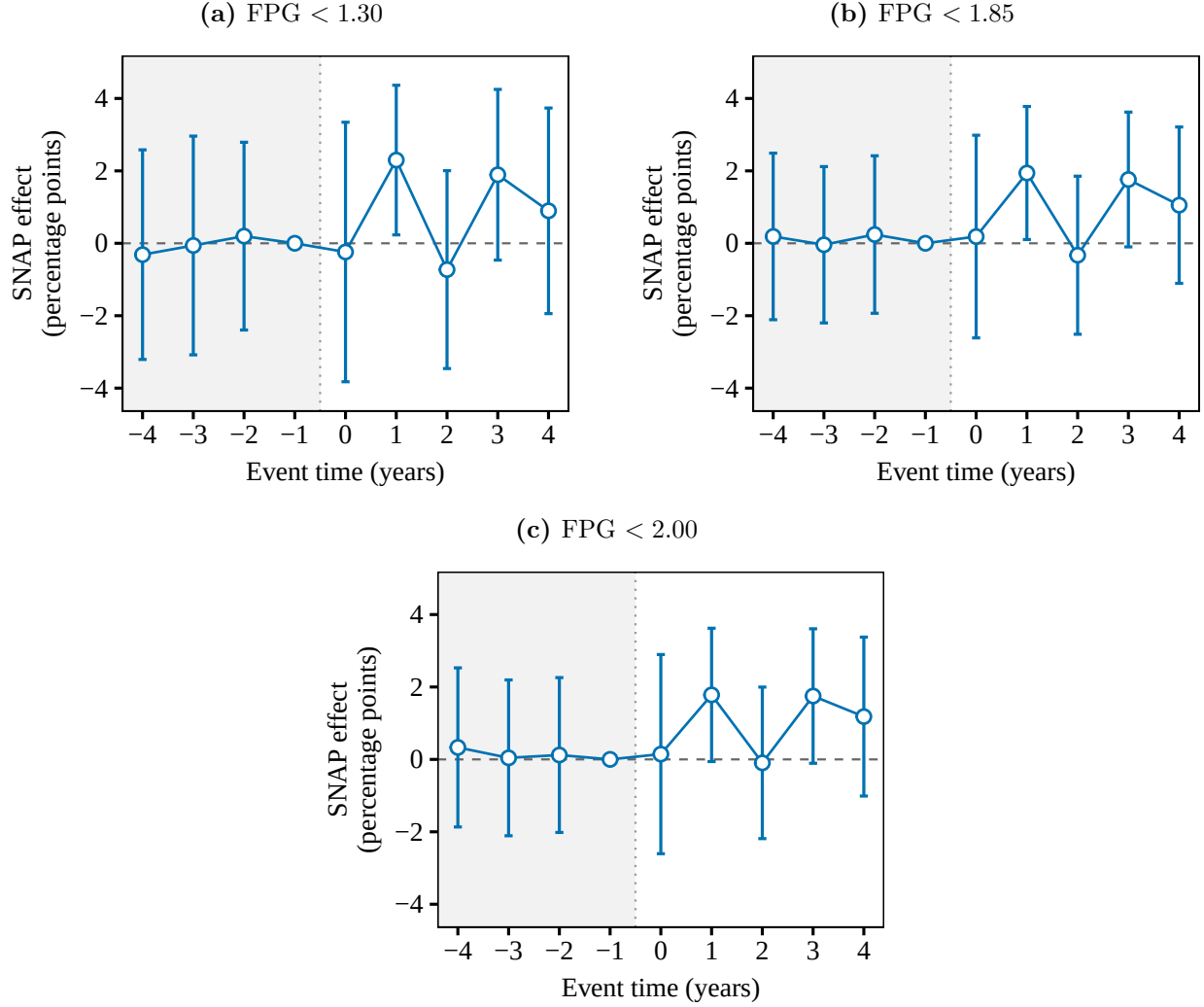
*Notes:* Dynamic coefficients from the clean stacked event study of within-state effective-minimum-wage increases of at least \$1.00 (21 clean events). The outcome is the household SNAP-receipt indicator copied to each person record; the sample is all individuals in the CPS-ASEC, income years 2000–2019. Estimates are shown in percentage points. The specification uses ASEC weights, individual demographic controls, house-price and Bartik labor-demand controls, and event-by-state and event-by-year fixed effects. The year before the increase ( $k = -1$ ) is normalized to zero. Shading marks the pre-event years; capped bars are 95% confidence intervals from state-clustered standard errors (51 states). The joint pre-trend  $p$ -value ( $k = -2, -3, -4$ ) is 0.40.

**Figure 4:** Clean Stacked Event Studies of SNAP Take-Up, by Subgroup



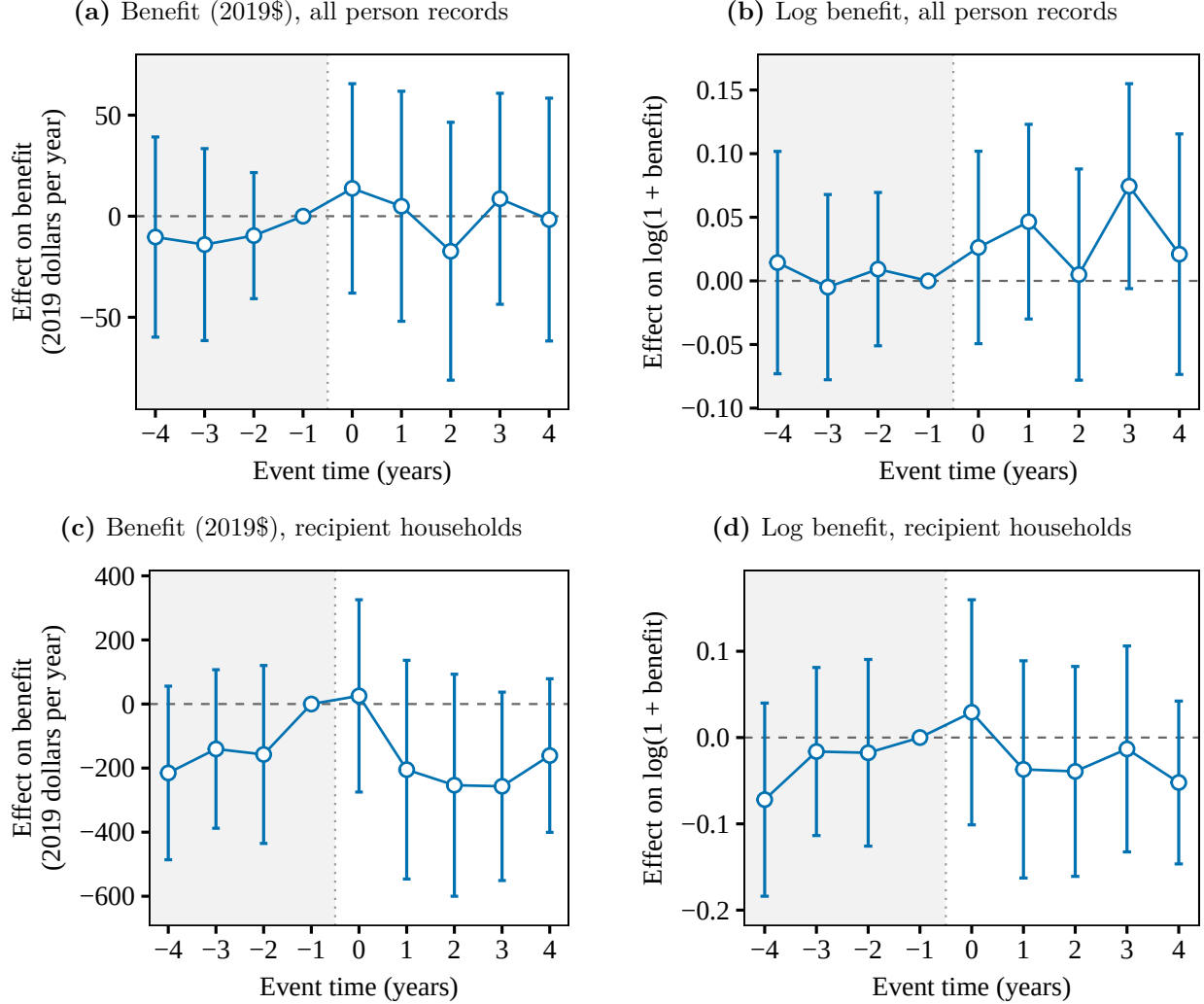
*Notes:* Estimates are shown in percentage points. Shading marks the pre-event years;  $k = -1$  is normalized to zero and shown without an interval. Each panel plots dynamic coefficients from the clean stacked event study of SNAP take-up on minimum-wage increases of at least \$1.00 (21 clean events), estimated on the individual micro data for the indicated subgroup, with the year before the increase ( $k = -1$ ) as the omitted reference, individual demographic controls, and house-price and Bartik labor-demand controls. The outcome is the household SNAP-receipt indicator copied to each person record. The sample is all individuals in the CPS-ASEC, income years 2000-2019; subgroups are defined over person records. Bars are 95% confidence intervals from state-clustered standard errors (51 states). Joint pre-trend  $p$ -values ( $k = -2, -3, -4$ ) by panel: working age 0.58, less educated 0.69, younger less educated 0.69, Black 0.20, Latino 0.02, immigrant (foreign-born) 0.62.

**Figure 5:** OLS Event Studies in FPG Demographic-Screen Samples



*Notes:* Estimates are shown in percentage points. Shading marks the pre-event years;  $k = -1$  is normalized to zero and shown without an interval. Dynamic OLS coefficients for the household SNAP-receipt indicator copied to each selected person record, from clean stacked event studies using the 20-event demographic-screen sample, with  $k = -1$  omitted. Panels (a)-(c) target FPG ratios below 1.30, 1.85, and 2.00. Each prediction model is estimated on pre-event income-target indicators and applied to observed row demographics throughout the window. Outcome regressions use ASEC weights and include house-price and Bartik controls and event-by-state and event-by-year fixed effects, but no individual demographic controls. Bars are 95% confidence intervals from state-clustered standard errors. Pooled estimates are about 0.8 percentage points and statistically insignificant ( $p = 0.25, 0.20$ , and  $0.18$ ). Joint pre-trend  $p$ -values are 0.90, 0.96, and 0.94. At  $k = +1$ , the estimates are 2.3, 1.9, and 1.8 points ( $p = 0.030, 0.039$ , and  $0.058$ ). The selected person records are demographic proxies for low-income status, not determinations of statutory SNAP eligibility.

**Figure 6:** Clean Stacked Event Studies: SNAP Benefit Amounts



*Notes:* Each panel plots dynamic coefficients from the clean stacked event study on within-state effective-minimum-wage increases of at least \$1.00 (21 clean events), with the year before the increase ( $k = -1$ ) as the omitted reference, ASEC weights, house-price and Bartik labor-demand controls, and event-by-state and event-by-year fixed effects, with no individual demographic controls. The outcome is the household SNAP benefit amount for the prior twelve months, deflated to 2019 dollars by the CPI-U and copied to each person record, in dollars per year (left column) and as the log of one plus the benefit (right column). Panels (a) and (b) use all person records in the CPS-ASEC, income years 2000-2019, with zeros for non-recipient households (sample mean \$372 per person-year). Panels (c) and (d) retain person records in households with positive SNAP receipt (mean \$3,622 per year); because receipt can respond to treatment, these panels condition on a post-treatment outcome. Shading marks the pre-event years, and the reference year is shown at zero without an interval. Bars are 95% confidence intervals from state-clustered standard errors (51 states). Joint pre-trend  $p$ -values ( $k = -2, -3, -4$ ): 0.92 in panel (a), 0.54 in panel (b), 0.50 in panel (c), 0.25 in panel (d).

## Appendix A Additional Robustness Results

This appendix reports two robustness checks referenced in Section 5.4: the whole-population null estimated on state-by-year cells, and the clean stacked estimate with and without the state SNAP-policy index of Ganong and Liebman (2018) added to the controls.

The event list in Table A1 documents the design in Section 4.1. The appendix also reports the propensity-weighted robustness checks and contemporaneous-income diagnostic referenced in Section 5.5.

**Table A1:** Minimum-Wage Increase Events Used in the Stacked Design

| State          | Event year | Observed window | Annual effective MW (\$) |        |        | Demographic screen |
|----------------|------------|-----------------|--------------------------|--------|--------|--------------------|
|                |            |                 | Before                   | After  | Change |                    |
| Alaska         | 2003       | 2000–2007       | 5.650                    | 7.150  | 1.500  | No                 |
| Arizona        | 2007       | 2003–2011       | 5.150                    | 6.750  | 1.600  | Yes                |
| Arizona        | 2017       | 2013–2019       | 8.050                    | 10.000 | 1.950  | Yes                |
| Colorado       | 2007       | 2003–2011       | 5.150                    | 6.850  | 1.700  | Yes                |
| Illinois       | 2005       | 2001–2009       | 5.500                    | 6.500  | 1.000  | Yes                |
| Iowa           | 2008       | 2004–2012       | 5.941                    | 7.250  | 1.309  | Yes                |
| Maryland       | 2007       | 2003–2011       | 5.150                    | 6.150  | 1.000  | Yes                |
| Michigan       | 2007       | 2003–2011       | 5.604                    | 7.051  | 1.447  | Yes                |
| Missouri       | 2007       | 2003–2011       | 5.150                    | 6.500  | 1.350  | Yes                |
| Montana        | 2007       | 2003–2011       | 5.150                    | 6.150  | 1.000  | Yes                |
| Nebraska       | 2016       | 2012–2019       | 8.000                    | 9.000  | 1.000  | Yes                |
| New Hampshire  | 2008       | 2004–2012       | 5.676                    | 6.755  | 1.079  | Yes                |
| New Jersey     | 2006       | 2002–2010       | 5.402                    | 6.402  | 1.000  | Yes                |
| New Jersey     | 2014       | 2010–2018       | 7.250                    | 8.250  | 1.000  | Yes                |
| New Mexico     | 2008       | 2004–2012       | 5.459                    | 6.522  | 1.063  | Yes                |
| North Carolina | 2007       | 2003–2011       | 5.150                    | 6.150  | 1.000  | Yes                |
| Ohio           | 2007       | 2003–2011       | 5.150                    | 6.850  | 1.700  | Yes                |
| Pennsylvania   | 2007       | 2003–2011       | 5.150                    | 6.704  | 1.554  | Yes                |
| Rhode Island   | 2015       | 2011–2019       | 8.000                    | 9.000  | 1.000  | Yes                |
| South Dakota   | 2015       | 2011–2019       | 7.250                    | 8.500  | 1.250  | Yes                |
| Washington     | 2017       | 2013–2019       | 9.470                    | 11.000 | 1.530  | Yes                |

*Notes:* The table lists the 21 clean events in the population-wide design. An event is a within-state increase of at least \$1.00 in the annual effective nominal minimum wage, with no other qualifying increase in that state within four years on either side. The effective rate is the higher of the annual-average state and federal rates in the minimum-wage series of Vaghul and Zipperer (2016). The before and after columns refer to income years  $t - 1$  and  $t$ ; the window column reports the years actually observed within  $[t - 4, t + 4]$  and 2000–2019. The demographic-screen analysis uses the 20 events with four complete pre-event years, excluding Alaska 2003. The screen column indicates treated-event inclusion; the two implementations also differ in their comparison-state sets. In the main design, comparison states have no qualifying increase in the corresponding window.

**Table A2:** Additional Robustness Checks: State-Year Aggregation and SNAP Policy Rules

|                                                               | SNAP estimate<br>(1) | Pre-trend $p$<br>(2) | Observations<br>(3) |
|---------------------------------------------------------------|----------------------|----------------------|---------------------|
| <i>Panel A: State-by-year aggregation</i>                     |                      |                      |                     |
| State-year SNAP share                                         | 0.0030<br>(0.0034)   | 0.43                 | 7,018               |
| <i>Panel B: SNAP policy index (Ganong and Liebman (2018))</i> |                      |                      |                     |
| Baseline (headline controls)                                  | 0.0045<br>(0.0031)   | 0.40                 | 27,367,646          |
| + SNAP policy index                                           | 0.0048<br>(0.0032)   | 0.39                 | 27,367,646          |

*Notes:* Both panels re-estimate the population-wide effect of a  $\geq \$1$  minimum-wage increase on the household SNAP-receipt indicator copied to each person record using the clean stacked event-study design (21 clean events, event threshold \$1.00); the estimate column reports the pooled static post-versus-pre ATT with the state-clustered standard error in parentheses, and the pre-trend column the joint  $p$ -value on the event-time leads ( $k = -2, -3, -4$ ). *Panel A* collapses the individual CPS-ASEC data to state-by-year cells and uses the whole-population SNAP participation share as the outcome, population-weighted, with event-by-state and event-by-year fixed effects and no individual controls; observations are stacked state-by-year cells. The estimate matches the individual-level headline, so the whole-population null is not an artifact of the micro data or its weighting. *Panel B* re-estimates the headline whole-population specification (Table 2, column 1: individual demographic controls, house-price and Bartik labor-demand controls, event-by-state and event-by-year fixed effects, 27,367,646 stacked person-years) with and without the state SNAP-policy index added to the state controls. The baseline row reproduces the headline estimate (0.0045); adding the index shifts it by 0.0003, well within its standard error. The index covers all sample years, so both rows are estimated on the identical sample. Standard errors are clustered by state (51 states).  $*p < 0.10$ ,  $**p < 0.05$ ,  $***p < 0.01$ .

**Table A3:** IPW Estimates in FPG Demographic Screens

| FPG benchmark | IPW<br>(1)         |
|---------------|--------------------|
| FPG < 1.30    | 0.0106<br>(0.0083) |
| FPG < 1.85    | 0.0102<br>(0.0070) |
| FPG < 2.00    | 0.0101<br>(0.0067) |

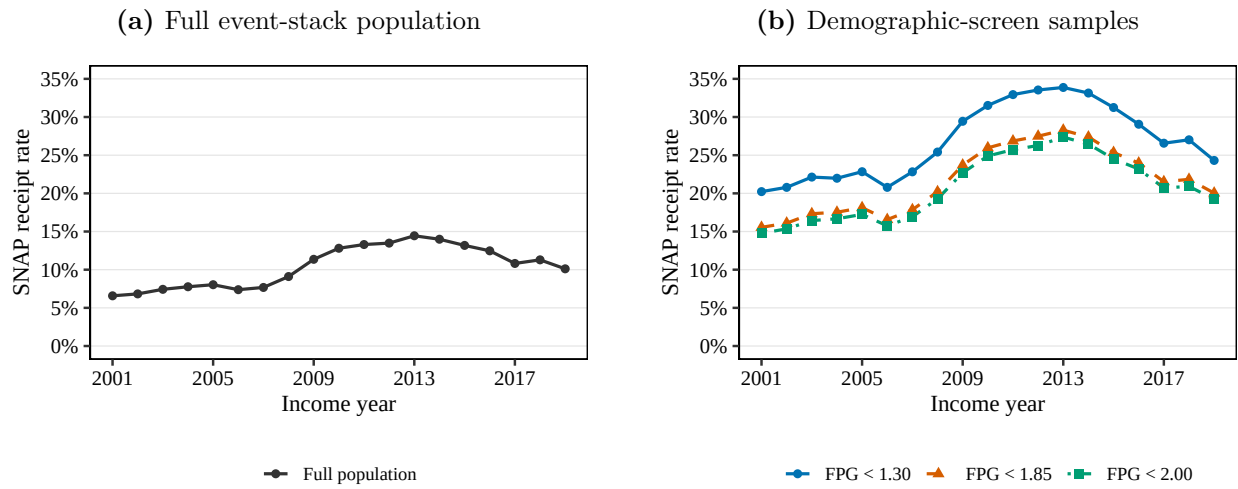
*Notes:* Static treatment-propensity-weighted stacked event-study estimates of the pooled post-versus-pre effect of a  $\geq$  \$1 minimum-wage increase on the household SNAP-receipt indicator copied to each selected person record. The demographic screens and outcome specification are those of Table 5. After screening, a separate additive event-specific logit of treated-state status uses age, sex, marital status, household size, presence of a child under six, race and ethnicity, and nativity to reweight control records toward treated records. Parentheses report standard errors from a 300-replication state-cluster bootstrap that re-estimates treatment weights while holding the demographic screen fixed. The corresponding sign-share bootstrap  $p$ -values are 0.19, 0.15, and 0.14. All estimates use 20 events.

**Table A4:** Contemporaneous Income-Screen Estimates

| Income target | OLS<br>(1)           | IPW<br>(2)            |
|---------------|----------------------|-----------------------|
| FPG < 1.30    | 0.0193**<br>(0.0092) | 0.0211***<br>(0.0071) |
| FPG < 1.85    | 0.0147*<br>(0.0074)  | 0.0167***<br>(0.0056) |
| FPG < 2.00    | 0.0140*<br>(0.0073)  | 0.0161***<br>(0.0054) |

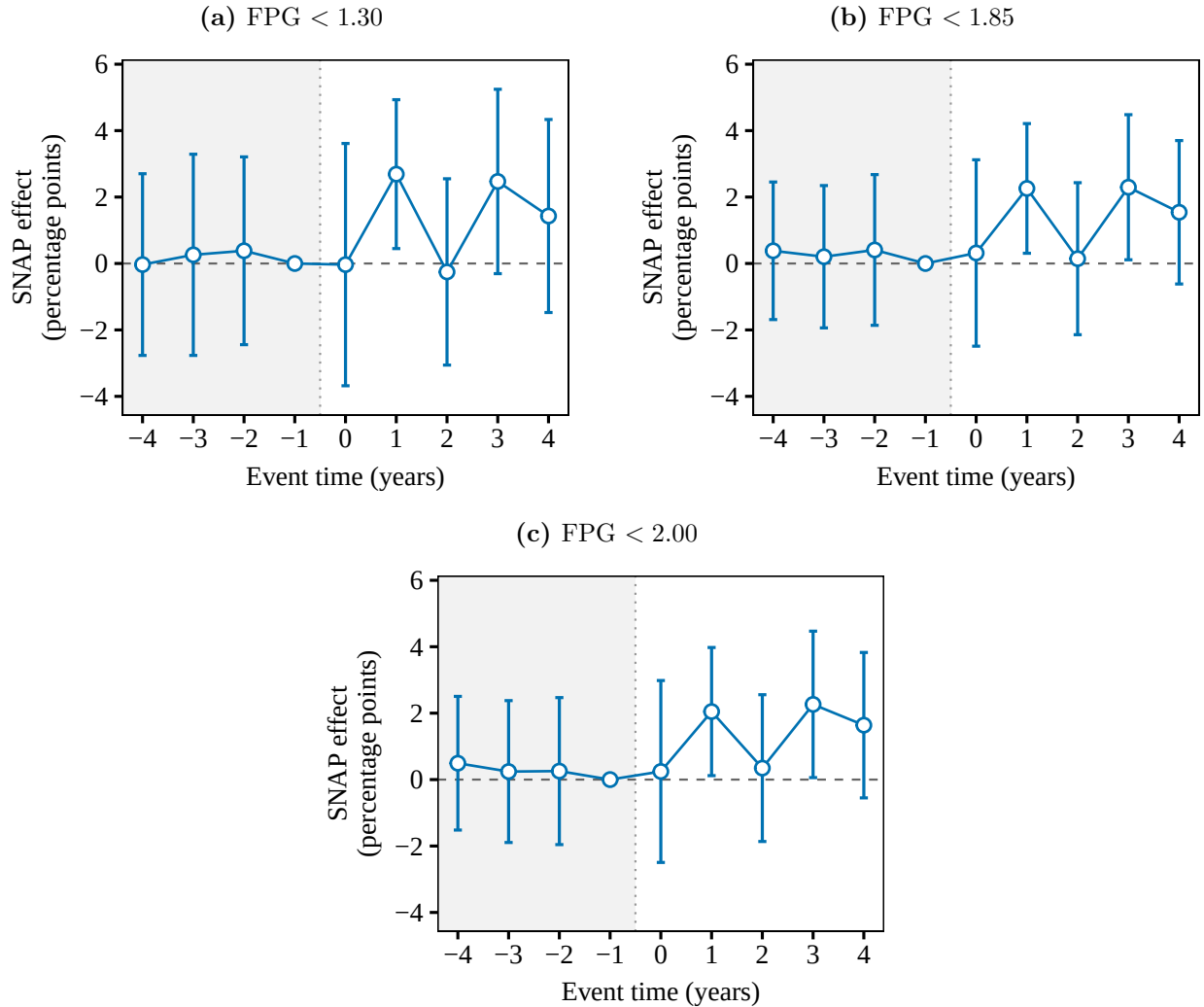
*Notes:* Static stacked event-study estimates of the pooled post-versus-pre effect of a  $\geq$  \$1 minimum-wage increase on the household SNAP-receipt indicator copied to each selected person record. The 20-event sample selects records using same-year family income, which is potentially affected by treatment. FPG denotes family income relative to the HHS Federal Poverty Guidelines. OLS uses ASEC weights. IPW additionally fits the separate additive event-specific treatment-status logit described in Section 5.5. Outcome regressions include house-price and Bartik controls and event-by-state and event-by-year fixed effects, but no additional individual demographic controls. Standard errors are state-clustered plug-in estimates. These estimates are a post-treatment-selection diagnostic, not a causal alternative to the demographic-screen estimates. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Figure A1:** Reported SNAP Receipt in the Full Event-Stack Population and Demographic-Screen Samples



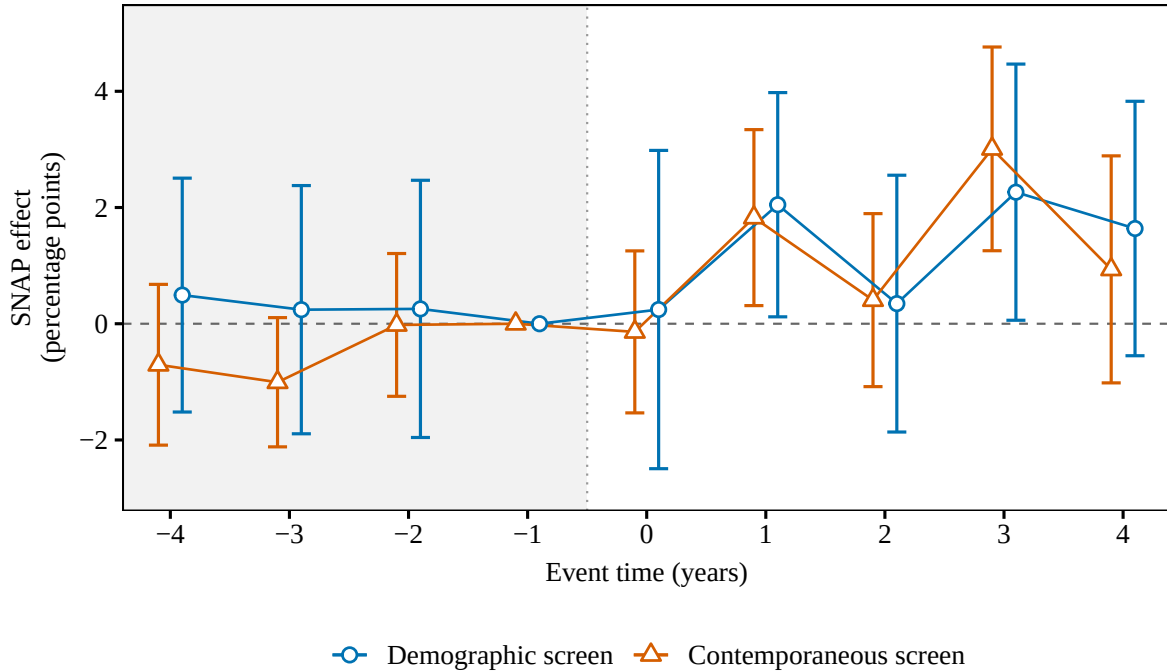
*Notes:* Annual ASEC-weighted shares of person-event records whose household reports SNAP receipt, using the CPS-ASEC and the 20 event windows retained in the demographic-screen analysis. Panel (a) uses the event-stack population before screening; panel (b) applies the event-specific demographic screens targeting family FPG ratios below 1.30, 1.85, and 2.00, as described in Section 5.5. Both panels pool treated and comparison states from the same event blocks and use a common vertical scale. Each annual share divides the sum of ASEC weights for records reporting receipt by the sum of ASEC weights for all records in that series and year; no treatment-propensity weights are applied. A person record appearing in multiple event windows contributes once per window and can satisfy a screen in one window but not another. The set of contributing events changes across income years. These are descriptive shares in stacked samples, rather than national annual participation rates, statutory-eligibility measures, or treatment-effect estimates.

**Figure A2:** Treatment-Propensity-Weighted Event Studies in FPG Demographic-Screen Samples



*Notes:* Estimates are shown in percentage points. Shading marks the pre-event years;  $k = -1$  is normalized to zero and shown without an interval. Dynamic treatment-propensity-weighted coefficients for the household SNAP-receipt indicator copied to each selected person record, using the 20-event demographic-screen sample with  $k = -1$  omitted. Panels (a)-(c) target FPG ratios below 1.30, 1.85, and 2.00. The demographic screens, treatment-propensity model, and outcome specification are those described in Section 5.5 and table A3. Bars are symmetric 95% confidence intervals formed from 300 state-cluster bootstrap draws; text uses the corresponding sign-share bootstrap  $p$ -values. Pooled estimates are about 1.0 percentage point and statistically insignificant ( $p = 0.19, 0.15$ , and  $0.14$ ). At  $k = +1$ , the estimates are 2.7, 2.3, and 2.0 points ( $p = 0.027, 0.033$ , and  $0.053$ ); at  $k = +3$ , they are 2.5, 2.3, and 2.3 points ( $p = 0.15, 0.060$ , and  $0.067$ ).

**Figure A3:** Demographic-Screen Event Study vs. Contemporaneous Income Screen (FPG < 2.00)



*Notes:* Estimates are shown in percentage points. Shading marks the pre-event years;  $k = -1$  is normalized to zero and shown without an interval. Dynamic coefficients for the household SNAP-receipt indicator copied to each selected person record, from propensity-weighted clean stacked event studies with  $k = -1$  omitted. The event-window series uses the 20-event demographic screen for the broad FPG target (FPG < 2.00): the prediction model is estimated on pre-event income-target indicators and applied to observed row demographics throughout the window. The contemporaneous series selects on same-year family income, a post-treatment variable. Outcome regressions include house-price and Bartik controls and event-by-state and event-by-year fixed effects, but no individual demographic controls. Event-window bars are 95% confidence intervals from a 300-replication state-cluster bootstrap that re-estimates treatment weights while holding the demographic screen fixed; contemporaneous bars use plug-in standard errors. Under the demographic screen, the pooled estimate is insignificant ( $p = 0.14$ ); the  $k = +1$  estimate is 2.0 points ( $p = 0.053$ ) and the  $k = +3$  estimate is 2.3 points ( $p = 0.067$ ).